

A NINE-UNIT FIBER FLOOR IN A FULL THREE-COVER BRANCH OF FRANKL'S CONJECTURE

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ABSTRACT. We derive exact restrictions inside the full minimal three-cover branch of the current structure theory for a smallest empty-set-free height-five counterexample to Frankl's union-closed sets conjecture. If q_0 is the size of the fibre with empty private support and q_2 counts the excess members in the three two-private-coordinate fibres, then

$$q_0 + q_2 \geq 9.$$

The proof combines deletion-block injections, exact source-capacity saturation, cross-fibre complement intersection, and height-three generation. Its last case uses a new near-extremal lemma: an intersection-closed height-three family generated by five nonempty sets and having at least fifteen members is the empty set, five pairwise-disjoint generators, and nine or ten pairwise unions. The lemma is proved by a ground-independent reduction to B_5 . Twenty fail-closed exact checks, implemented twice at every proof-essential finite reduction, audit the boundary-eight case split and the Boolean classification. The result is strictly conditional: it does not eliminate the full three-cover branch, the four-cover branches, or the general height-five case. At the first residual size $r = 10$, for $t = 27$, the two independent bounds collapse the fibre parameters to the unique triple $(q_0, u, q_2) = (4, 0, 6)$.

1. INTRODUCTION

Frankl's union-closed sets conjecture asserts that every finite nontrivial union-closed family has an element contained in at least half of its members. We refer to [1, 3] for background. The height of a family is the largest number of members in a strict inclusion chain. Chain conditions have proved useful in this problem: Tian established the height-at-most-three case [4], and Colbert subsequently obtained a general chain-condition proof in a broader setting [2].

Recent work of Tian proves the empty-set-free height-at-most-four case and develops a conditional structure theory for the next height [6]. In the normalization used there, a smallest hypothetical height-five counterexample has $2t$ members, critical coordinates of frequency t , and a critical join-cover

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number equal to three or four. The remaining branches are explicitly left open.

This note treats the full three-cover branch at general residual parameters and then sharpens one exact parameter point. Let B_1, B_2, B_3 be a full minimal critical join-cover and choose private critical coordinates $x_i \in B_i \setminus \bigcup_{j \neq i} B_j$. Put

$$X = \{x_1, x_2, x_3\}, \quad R = U \setminus X, \quad r = |R|.$$

For $I \subseteq [3]$, the I -fiber consists of members whose intersection with X is $\{x_i : i \in I\}$. The seven nonempty Boolean unions of the B_i give one distinguished member in every nonempty fiber. Let q_I count the remaining members in that fiber, and let q_0 denote the size of the empty-support fiber. Our first main result is uniform.

Remark 1.1 (Provenance firewall). The smallest-counterexample normalization, critical-coatom theorem, critical-pair dichotomy, nonfull three-trace normal form and layer counts, projection-witness inequality, and the restriction $\rho_c \in \{3, 4\}$ are imported from Tian [6]. The assumption here that $\rho_c = 3$ admits a *full* minimal cover is a conditioning choice inside one branch that Tian leaves open, not a theorem of that source. The lower cutoff $r \geq 10$ uses the independent computer-assisted verification of Frankl's conjecture on ground sets of size at most twelve [5]; it is not imported from Tian. The three-cover fiber equations and heights, complement blocks, matching caps, exact-threshold lemma, block-projection inequalities, boundary exclusions, the height-three generator bound, and Lemmas 3.5 and 3.7 are derived and proved here. In particular, the singleton-fiber height bound used in those lemmas follows from the chain-extension argument in this note and is not an additional Tian import. The source package includes a theorem-by-theorem and equation-by-equation import crosswalk.

Theorem 1.2. *In the full three-cover branch of a smallest empty-set-free height-five counterexample described in [6], write*

$$r = |R| \geq 10, \quad t = 2r + 5 + s, \quad s \geq 0.$$

Then

$$q_0 \leq \left\lceil \frac{2r + s}{3} \right\rceil - 1.$$

Since the three private coordinates give $|U| = r + 3$, the verified twelve-element case rules out $r \leq 9$. Thus $r = 10$ is the first residual size not covered by that finite-ground result. At this first remaining value, the same projection mechanism can be retained without the ground-independent relaxations. Indeed, adjoining $\emptyset$ to an empty-set-free, $2t$ -member strict-majority counterexample preserves both its ground set and all coordinate frequencies. The resulting $(2t + 1)$ -member family would have no coordinate in at least $t + 1$ members and hence would be an ordinary Frankl counterexample, so the verified twelve-element result applies in the present normalization.

Theorem 1.3. *Under the hypotheses of Theorem 1.2, if $r = 10$ and $t = 27$ (equivalently, $s = 2$), then $q_0 \leq 4$.*

Our projection-matching result controls the opposite end of the parameter range. Put

$$q_2 = q_{12} + q_{13} + q_{23}.$$

Theorem 1.4. *Under the hypotheses of Theorem 1.2,*

$$q_0 + q_2 \geq 2.$$

Equivalently, in the parametrization of Lemma 2.1,

$$2q_0 + s - 2u \geq 2.$$

The equality boundary has additional rigidity. For each private source let $c_i = |\mathcal{J}_i|$ be its exact deletion-matching size, and put $H = \{i : c_i = 5\}$.

Theorem 1.5. *If $q_0 + q_2 = 2$, then $|H| \leq 2$. If $(q_0, q_2) = (1, 1)$ or $(2, 0)$, then $|H| \leq 1$.*

If $(q_0, q_2) = (0, 2)$ and $H = \{a, b\}$, with h the remaining index, then

$$q_{ab} = 2, \quad q_{ah} = q_{bh} = 0.$$

The row h has $c_h = 3$, has a residual critical witness z , and the three members of the ab -fiber form a strict chain

$$E < T_{h,z} < C_h.$$

If $(q_0, q_2) = (2, 0)$ and $H = \{i\}$, the maximum of the support-zero fiber is R ; the other two matching rows have size three and use R as their unique support-zero lower endpoint.

The boundary classification combines with the exact nonfull-pair trace formula to improve the preliminary floor by one.

Theorem 1.6. *Under the hypotheses of Theorem 1.2,*

$$q_0 + q_2 \geq 3.$$

Equivalently,

$$2q_0 + s - 2u \geq 3.$$

The next three boundaries can also be removed, but require cross-fiber complement intersection and height-three generation rather than matching-row arithmetic alone.

Proposition 1.7. *Under the hypotheses of Theorem 1.2,*

$$q_0 + q_2 \geq 6.$$

Theorem 1.8. *Under the hypotheses of Theorem 1.2,*

$$q_0 + q_2 \geq 9.$$

Equivalently,

$$2q_0 + s - 2u \geq 9.$$

Corollary 1.9 (Collapse at $(r, t) = (10, 27)$). *Under the same hypotheses, if $r = 10$ and $t = 27$, then*

$$(q_0, u, q_2) = (4, 0, 6), \quad q_0 + q_2 = 10.$$

Proof. Here $s = 2$. Theorem 1.3 gives $q_0 \leq 4$, while Theorem 1.8 and the balance identity $q_2 = q_0 + s - 2u$ give $2q_0 + 2 - 2u \geq 9$, hence $q_0 - u \geq 4$. Since $u \geq 0$, equality is forced: $q_0 = 4, u = 0$, and then $q_2 = 6$. $\square$

Consequently the two independent mechanisms confine every configuration to

$$(1) \quad 9 - q_2 \leq q_0 \leq \left\lceil \frac{2r + s}{3} \right\rceil - 1.$$

At $(r, t) = (10, 27)$ this window collapses completely, as stated in Corollary 1.9; this is stronger than either mechanism in isolation.

Theorem 1.2 follows from explicit inequalities. Theorem 1.3 uses the exact block-projection recurrence in Section 6; its finite evaluation uses integer arithmetic. Theorem 1.4 uses exact projection matchings and is independent of the block-incidence argument. Theorem 1.5 analyzes equality in the resulting matching cap and preserves residual traces throughout. Theorem 1.6 then excludes that equality boundary using Tian's exact three-trace formula. Theorem 1.8 combines threshold rigidity, complement intersection, coordinate-omission capacity, height-three generation, deletion-block projection, and four- and five-generator cross-meet lemmas on the next six boundaries. Section 9 records independent verification strategies. These theorems do not eliminate the full three-cover branch or prove Frankl's conjecture at height five. The next unresolved lower boundary is $q_0 + q_2 = 9$.

2. FIBER EQUATIONS AND COMPLEMENT BLOCKS

We recall only the consequences of the smallest-counterexample structure needed below. The family is separating and union-closed, every coordinate has frequency at most t , and the three private coordinates have frequency exactly t . Write $t = 2r + 5 + s$ and put

$$d = t - 4 = 2r + 1 + s.$$

For a coordinate z , let

$$o(z) := |\{F \in \mathcal{F} : z \notin F\}|$$

be its omission count. Since $|\mathcal{F}| = 2t$ and every coordinate has frequency at most t , one has $o(z) \geq t$ for every $z \in U$.

For $j = 0, 1, 2, 3$, write

$$q_j = \sum_{|I|=j} q_I.$$

Lemma 2.1 (Fiber balance). *There is an integer $u \geq 0$ such that*

$$(2) \quad q_3 = r + u, \quad q_2 = q_0 + s - 2u, \quad q_1 = 3r + s + u + 3 - 2q_0.$$

If x_{12}, x_{13}, x_{23} are the three pair-fiber counts and

$$w = r + 1 + s - u,$$

then their sum is $q_0 + s - 2u$ and the three singleton-fiber counts are

$$(3) \quad w - x_{12} - x_{13}, \quad w - x_{12} - x_{23}, \quad w - x_{13} - x_{23}.$$

All six numbers in (3) are nonnegative and at most w . Moreover, the singleton counts are at least $\ell := r + 1 - q_0 + u$, and the sum of all six counts is $2w + \ell$.

Proof. There are $2t$ members and seven nonempty Boolean members, so $q_0 + q_1 + q_2 + q_3 = 2t - 7 = 4r + 2s + 3$. Counting the three private coordinates among the non-Boolean members gives

$$q_1 + 2q_2 + 3q_3 = 3d.$$

For every $y \in R$, the maximal member avoiding y contains all of X ; these r members are distinct. Thus $q_3 \geq r$, and we write $q_3 = r + u$. Solving the two displayed equations gives (2). Finally, each private coordinate occurs in d non-Boolean members. After removing the q_3 full-support members this gives the three equations in (3). The upper bound by w is immediate. Each singleton count is at least $w - q_2 = \ell$, and summation gives $2w + \ell$. $\square$

Let $\mathcal{G}$ be the empty-support fiber. It is union-closed and has height at most two: any chain in $\mathcal{G}$ can be followed by the three successive joins with B_1, B_2, B_3 . Let M be its maximum and put $m = |M|$. If $q_0 = q > 0$, write its other members as

$$M \setminus D_1, \dots, M \setminus D_{q-1}.$$

Lemma 2.2 (Complement blocks). *The nonempty sets $D_1, \dots, D_{q-1}$ are pairwise disjoint. In particular,*

$$q_0 - 1 \leq m \leq r,$$

so $q_0 \leq r + 1$.

Proof. The proper members of a height-two union-closed family are incomparable. Hence the union of two distinct proper members must be M . Taking complements in M gives $D_i \cap D_j = \emptyset$. The cardinality bound follows. $\square$

Write

$$(4) \quad d_i = |D_i|, \quad S = \sum_i d_i, \quad h = m - S.$$

The h coordinates are called uncovered coordinates of M .

3. PROJECTION MATCHINGS AND THE LOW-END FLOOR

For a private critical coordinate x_i , define its exact deletion matching

$$\mathcal{J}_i = \{A \in \mathcal{F} : x_i \notin A, A \cup \{x_i\} \in \mathcal{F}\}, \quad c_i = |\mathcal{J}_i|.$$

For $z \neq x_i$, put

$$j_i(z) = |\{A \in \mathcal{J}_i : z \in A\}|, \quad \delta_z = t - |\{F \in \mathcal{F} : z \in F\}|,$$

and $D_i(z) = c_i - 2j_i(z) - 2\delta_z$.

We use the following exact consequences of the projection part of Tian's smallest-counterexample reduction. They are included here to fix all equality cases needed below.

Lemma 3.1 (Small matching rows). *For every i , the family $\mathcal{J}_i$ is union-closed, contains $C_i = U \setminus \{x_i\}$, and satisfies $c_i \geq 3$. Some $z \neq x_i$ has $D_i(z) \geq 1$. If $c_i \in \{3, 4\}$, there is a unique such positive-defect coordinate; it is critical, and for it $j_i(z) = 1$. Moreover,*

$$(5) \quad T_{i,z} := U \setminus \{x_i, z\} \in \mathcal{J}_i,$$

and $T_{i,z}$ is the maximum of $\mathcal{J}_i \setminus \{C_i\}$.

Proof. Exact deletion of x_i shows that $\mathcal{J}_i$ is union-closed. The coatom-top edge gives $C_i \in \mathcal{J}_i$. The projection comparison in the minimal counterexample gives a positive defect $D_i(z) \geq 1$; its small-row form gives

$$(6) \quad c_i \geq 2\delta_z + 3, \quad j_i(z) \geq 1.$$

It also gives $c_i \geq 3$. Thus, when c_i is three or four, every positive witness is critical and has $j_i(z) = 1$.

For completeness, uniqueness follows from a short moment count. If W is the set of critical positive witnesses, then every pair (x_i, z) with $z \in W$ is full. Indeed, if that pair were nonfull, Tian's projection localization inside a nonfull pair would place every projection witness for x_i outside the exceptional coordinate set containing x_i and z ; thus z itself could not be a witness. Pair fullness, together with the two critical coatoms C_i and C_z , gives (5). The members C_i and $T_{i,z}$ for $z \in W$ have total incidence at least $|W|^2$ on W . Consequently

$$\sum_{z \in W} D_i(z) = |W|c_i - 2 \sum_{z \in W} j_i(z) \leq |W|(c_i - 2|W|).$$

The left side is at least $|W|$, so $|W| \leq \lfloor (c_i - 1)/2 \rfloor = 1$. Existence gives uniqueness. Finally, C_i is the only lower endpoint containing z . Every other endpoint is therefore contained in their union, which is $T_{i,z}$ by union closure and (5). $\square$

If $\{i, j, k\} = [3]$, the four possible private supports of a lower endpoint in $\mathcal{J}_i$ are

$$\emptyset, \quad \{j\}, \quad \{k\}, \quad \{j, k\}.$$

Exact addition of x_i is injective. Counting the four support classes against their target fibers gives the useful cap

$$(7) \quad c_i \leq q_0 + (1 + q_{ij}) + (1 + q_{ik}) + (1 + q_{jk}) = q_0 + q_2 + 3.$$

Proof of Theorem 1.4. Suppose first that $q_0 = q_2 = 0$. Equations (7) and $c_i \geq 3$ give $c_i = 3$. Let z_i be the unique witness from Lemma 3.1 and put $T_i = T_{i,z_i}$. Choose $A < T_i$ below the coatom. If z_i is residual, then $T_i < C_i$ are two members in the same pair fiber. If $z_i = x_j$ and k is the third private index, then $A < T_i$ have singleton support $\{k\}$, and their distinct x_i -extensions lie in one pair fiber. Both alternatives force $q_2 \geq 1$, a contradiction.

Next suppose $q_0 = 0$ and $q_2 = 1$. Let E be the unique pair support with an additional member. By (7), every c_i is three or four. If the witness in row i is residual, $T_i < C_i$ force $E = [3] \setminus \{i\}$. If it is x_j , the same extension argument forces $E = [3] \setminus \{j\}$. Write $E = [3] \setminus \{h\}$. The row h therefore has a residual witness z_h , and its exceptional pair fiber is exhausted by $T_h < C_h$. Choose $A \in \mathcal{J}_h \setminus \{T_h, C_h\}$. Since $A < T_h$, absence of an empty-support member and exhaustion of E force A to have singleton support $\{a\}$ for some $a \in E$. The upper endpoint $A \cup \{x_h\}$ lies in a nonexceptional pair fiber, so it is the unique Boolean member there, namely C_b for the other $b \in E$. This is impossible: $A \cup \{x_h\}$ omits z_h , whereas C_b contains z_h .

It remains to exclude $q_0 = 1, q_2 = 0$. A residual witness would create a second member in a pair fiber, so each unique witness is private. In every row, a member strictly below T_i must be the unique support-zero member Z ; otherwise its two distinct extensions again create an additional pair member. Hence $Z, Z \cup \{x_i\} \in \mathcal{F}$ for all i . Union closure and pair-fiber uniqueness give

$$Z \cup \{x_i, x_j\} = C_k \quad (\{i, j, k\} = [3]),$$

so $Z = R$. For fixed i , the four distinct lower endpoints

$$Z, \quad Z \cup \{x_j\}, \quad Z \cup \{x_k\}, \quad C_i$$

belong to $\mathcal{J}_i$. Equation (7) makes these all of them. Every residual coordinate has matching frequency four, while x_j, x_k have frequency two. Thus $D_i(z) \leq 0$ for every $z \neq x_i$, contradicting Lemma 3.1. All cases with $q_0 + q_2 \leq 1$ are impossible. $\square$

We next analyze equality at the first surviving boundary. Equality in (7) is equality separately in every private support class; it does not identify unrelated residual traces.

Lemma 3.2 (Five-row equality). *Assume $q_0 + q_2 = 2$ and $c_i = 5$. Every support-zero member is a lower i -matching endpoint. Every member of a pair fiber containing i has its x_i -deletion in $\mathcal{F}$, and every member of the pair fiber not containing i is a lower i -matching endpoint. No residual coordinate has positive i -defect. If $\{i, j, k\} = [3]$, then*

$$(8) \quad D_i(x_j) = 1 - 2q_2 + 2q_{ik}.$$

Proof. The four terms on the right of (7) bound four disjoint support classes. Since their sum equals $c_i = 5$, every class bound is an equality. Exact addition of x_i gives the stated literal deletions, so every residual trace is preserved.

The lower matching family contains C_i and the two deletions

$$C_k \setminus \{x_i\} = R \cup \{x_j\}, \quad C_j \setminus \{x_i\} = R \cup \{x_k\}.$$

Thus every residual coordinate occurs in at least three of five lower endpoints and has defect at most -1 . Finally,

$$j_i(x_j) = (1 + q_{ij}) + (1 + q_{jk}) = 2 + q_2 - q_{ik},$$

which gives (8) because x_j is critical. $\square$

Proof of Theorem 1.5. First let $(q_0, q_2) = (0, 2)$. By (8), a five-row is possible only when both units of pair mass lie in one pair incident with its source. Hence at most the two endpoints of that pair are high. If they are a, b , then $q_{ab} = 2$ and the other pair counts vanish. The complementary row h is low. Lemma 3.1 gives it a unique critical witness. The only occupied pair fiber is ab , so the witness is residual and $T_{h,z} < C_h$ lie in that fiber. A singleton lower endpoint would extend to a nonexceptional pair coatom; its deletion contains z , contradicting $j_h(z) = 1$. Support zero is absent. Thus all of $\mathcal{J}_h$ lies in the three-member ab -fiber. Since $c_h \geq 3$, it has size three, and the maximal property in Lemma 3.1 gives $E < T_{h,z} < C_h$.

Now let $(q_0, q_2) = (1, 1)$. Equation (8) confines every high source to an endpoint of the exceptional pair $\{a, b\}$. Suppose both endpoints were high and let h be the remaining source. Its unique witness is residual, and $T_{h,z} < C_h$ exhaust the exceptional pair fiber. If Z is the unique support-zero member, high-row equality gives

$$(9) \quad Q := Z \cup \{x_a, x_b\} \in \{T_{h,z}, C_h\}.$$

Choose $A < T_{h,z}$ in $\mathcal{J}_h$. It cannot have exceptional pair support, which is exhausted, or singleton support, whose extension would be a nonexceptional coatom containing z . Hence $A = Z$.

If $Q = C_h$, then $Z = R$, which is not contained in $T_{h,z}$. If $Q = T_{h,z}$, then $Z = R \setminus \{z\}$. But Z is matched in rows h and a , so their two extensions have union in a nonexceptional pair fiber; uniqueness there forces $Z = R$, again a contradiction. Therefore at most one row is high.

Finally let $(q_0, q_2) = (2, 0)$, with support-zero members $A < M$. If two rows i, j were high, both A and M would be matched in both directions. For either $Z \in \{A, M\}$, union closure places $Z \cup \{x_i, x_j\}$ in the unique ij -fiber, whose member is C_k . Thus both A and M would equal R , impossible. Hence at most one row is high.

Suppose it is row i . Both support-zero members are matched in direction i . A low row cannot have a residual witness, since its canonical pair would create a second member in a pair fiber. A member below its canonical singleton endpoint must have support zero; a singleton alternative would

again create an extra pair member. Such a support-zero member is matched in its low direction and in direction i , so pair-fiber uniqueness forces it to be R . Both low rows therefore use $M = R$. A low row of size four would use both support-zero members and force $A = R$ as well. Hence both low rows have size three and the asserted endpoint pattern follows. $\square$

The preceding matching classification can be combined with the exact three-trace normal form for a nonfull critical pair.

Lemma 3.3 (The exact threshold is full). *If x, y are critical and*

$$d_{xy} = |\{F \in \mathcal{F} : x, y \notin F\}| = \left\lceil \frac{t+1}{2} \right\rceil,$$

then their double-avoidance top is $U \setminus \{x, y\}$.

Proof. Suppose instead that the pair is nonfull. In Tian's three-trace normal form, write P for the coordinates outside the double-avoidance top, $k = |P| \geq 3$, $g = d_{xy}$, and $\mathcal{W}$ for the family of full- P traces, with $w = |\mathcal{W}|$. The exact layer identity is

$$w = (k-1)g - (k-2)t - \sum_{p \in P} \delta_p.$$

If $t = 2m + 1$ is odd, substitution of $g = m + 1$ gives

$$w = (3-k)m + 1 - \sum_{p \in P} \delta_p.$$

If $t = 2m$ is even, substitution of $g = m + 1$ gives

$$w = (3-k)m + k - 1 - \sum_{p \in P} \delta_p.$$

Here $t \geq 2r + 5 \geq 25$. In either parity, $w \geq 1$ forces $k = 3$; the displayed formula then gives $w \leq 1$ in the odd case and $w \leq 2$ in the even case. Since $x \in P$ is critical, projection inside the same nonfull pair identifies its lower matching family with $\mathcal{H}_x \cap \mathcal{W}$. Consequently

$$c_x = |\mathcal{H}_x \cap \mathcal{W}| \leq |\mathcal{W}| = w \leq 2,$$

contradicting $c_x \geq 3$ from Lemma 3.1. $\square$

Proof of Theorem 1.6. Theorem 1.4 leaves only the possibility $q_0 + q_2 = 2$. For a private pair i, j , with remaining index k , direct fiber counting and Lemma 2.1 give

$$(10) \quad d_{ij} = q_0 + (1 + q_{\{k\}}) = r + 2 + u + q_{ij}.$$

On the boundary,

$$s = 2 - 2q_0 + 2u, \quad \frac{t+1}{2} = r + 4 - q_0 + u,$$

so

$$\frac{t+1}{2} - d_{ij} = 2 - q_0 - q_{ij} = q_2 - q_{ij} \geq 0.$$

The critical-pair dichotomy says that a nonfull pair must have $d_{ij} \geq (t+1)/2$. Equality would then hold, which Lemma 3.3 excludes. Thus all three private pairs are full.

Theorem 1.5 supplies a source i with $c_i \leq 4$. If $\{i, j, k\} = [3]$, fullness of the pairs (i, k) and (i, j) puts

$$R \cup \{x_j\}, \quad R \cup \{x_k\}$$

in $\mathcal{J}_i$; the coatom C_i is a third lower endpoint. Every residual coordinate therefore has $j_i(z) \geq 3$, while each of x_j, x_k has $j_i \geq 2$. Since $c_i \leq 4$, all their defects are nonpositive, contradicting the mandatory positive witness in Lemma 3.1. Hence the boundary is impossible and $q_0 + q_2 \geq 3$. $\square$

3.1. The next matching boundary. We now exclude equality in the preceding theorem. The first step records the common structure of all four possible distributions.

Lemma 3.4. *Suppose $q_0 + q_2 = 3$. Then all three private critical pairs are full. Every private matching row has size five or six, and every projection witness is one of the other two private coordinates. Moreover, $(q_0, q_2) = (0, 3)$ is impossible; if $(q_0, q_2) = (1, 2)$, the two units of pair-fiber mass occupy two distinct pair supports.*

Proof. For a private pair i, j , with remaining index k , the calculation in (10) gives $d_{ij} = r + 2 + u + q_{ij}$. On the present boundary,

$$s = 3 - 2q_0 + 2u, \quad t = 2r + 8 - 2q_0 + 2u,$$

and hence

$$\left\lceil \frac{t+1}{2} \right\rceil - d_{ij} = 3 - q_0 - q_{ij} = q_2 - q_{ij} \geq 0.$$

The critical-pair dichotomy and Lemma 3.3 therefore force every private pair to be full.

Fix a source i and write $\{i, j, k\} = [3]$. Pair fullness and the coatom give the three distinct lower endpoints

$$R \cup \{x_j\}, \quad R \cup \{x_k\}, \quad C_i$$

in $\mathcal{J}_i$. Since $c_i \leq q_0 + q_2 + 3 = 6$, no residual coordinate can be a projection witness, and if $c_i \leq 4$ no private coordinate can be one. Thus $c_i \in \{5, 6\}$ and a witness is private.

If x_j witnesses source i , the two displayed endpoints containing x_j force $j_i(x_j) = 2$. Every other lower endpoint avoids x_j , so its private support is empty or $\{k\}$. There are at most q_0 of the first kind; deletion injects the second kind into the ik -fiber, which has $1 + q_{ik}$ members. Consequently

$$(11) \quad 5 \leq c_i \leq 2 + q_0 + (1 + q_{ik}), \quad q_{ik} \geq 2 - q_0 = q_2 - 1.$$

Every vertex in the three-vertex pair-support graph must therefore meet an edge of weight at least $q_2 - 1$. Three units cannot give every vertex an incident edge of weight at least two, excluding $(0, 3)$. With two units,

covering all three vertices by positive edges forces weights $(1, 1, 0)$ up to permutation. $\square$

For the remaining cases, take complements on the residual ground set. For $I \subseteq [3]$, let $\mathcal{C}_I$ consist of $R \setminus (F \cap R)$ over members F in the I -fiber. Union closure gives

$$(12) \quad D \in \mathcal{C}_I, E \in \mathcal{C}_J \implies D \cap E \in \mathcal{C}_{I \cup J}.$$

Whenever the private pair complementary to i is full, the singleton- i fiber contains its full-residual top. Critical coatoms and U always give the corresponding empty residual complements in pair and full support. In the boundary-three discussion just completed all private pairs are full, so there every nonempty private fiber has the empty residual complement. Put $R_i = \bigcup \mathcal{C}_{\{i\}}$. In the boundary-four argument below we will invoke the empty-complement assertion only for the particular pairs already known to be full.

Lemma 3.5 (Four-generator cross-meet). *Let*

$$\mathcal{L} = \{\emptyset, D_1, D_2, D_3, D_4\}$$

be intersection-closed, where the D_i are distinct and nonempty, and let $\mathcal{U}$ consist of all unions of subcollections of the D_i . If $\mathcal{A}, \mathcal{B} \subseteq \mathcal{U}$ both contain $\emptyset$ and

$$A \cap B \in \mathcal{L} \quad (A \in \mathcal{A}, B \in \mathcal{B}),$$

then

$$\min\{|\mathcal{A}|, |\mathcal{B}|\} \leq 8.$$

Proof. The set relations among four generators are determined by the presence or absence of the fifteen nonzero Venn membership words in $\{0, 1\}^4$; atom multiplicity is irrelevant. Thus 2^{15} patterns exhaust arbitrary ground sets. For each pattern, retain it only when the four generators are nonempty and distinct and their five-element family is intersection-closed. Form its at-most-sixteen unions. On these unions put an edge $A \sim B$ when $A \cap B \in \mathcal{L}$. For every left vertex set containing $\emptyset$, intersect the exact neighbour sets of its vertices. This is the largest possible compatible right family.

The exact enumeration retains 301 labelled realizations. Its orbit profile under permutation of the four generators is

$ \mathcal{U} $	5	6	7	8	8	9	9	10	12	16
balanced bound	5	5	5	5	6	5	6	6	7	8
number of orbits	3	5	2	3	1	2	1	2	1	1.

The maximum is eight. A second enumeration starts instead from the 5^6 pair-meet tables on $\{0, D_1, D_2, D_3, D_4\}$, retains associative, commutative, idempotent operations, derives their allowed Venn words, and uses a recursive common-neighbour search. It finds fifteen abstract meet types, recovers the same 301 concrete realizations, and again returns eight. Both integer verifiers and their outputs are described in Section 9. $\square$

Lemma 3.6 (Height-three generator bound). *Let $\mathcal{C}$ be a finite intersection-closed family containing $\emptyset$ and having height at most three. If fixed nonempty sets $G_1, \dots, G_k$ generate every member of $\mathcal{C}$ by unions, then*

$$|\mathcal{C}| \leq 1 + k + \binom{k}{\lfloor k/2 \rfloor}.$$

For $k = 2, 3, 4, 5$ the respective bounds are 5, 7, 11, 16.

Proof. The assertion is immediate for $\mathcal{C} = \{\emptyset\}$, so suppose that $\mathcal{C}$ is non-trivial. For $A \in \mathcal{C}$ put

$$\sigma(A) = \{i \in [k] : G_i \subseteq A\}.$$

Because A is generated by a subcollection of the G_i , it equals the union of all G_i indexed by $\sigma(A)$. Thus σ is an injective order embedding and

$$\sigma(A \cap B) = \sigma(A) \cap \sigma(B).$$

The minimal nonempty members of $\sigma(\mathcal{C})$ are pairwise disjoint, so there are at most k of them. All remaining nonempty members form an antichain: comparable nonatoms $X \subsetneq Y$, together with an atom $P \subsetneq X$, would give the forbidden chain $\emptyset \subsetneq P \subsetneq X \subsetneq Y$. Sperner's theorem bounds the nonatoms by $\binom{k}{\lfloor k/2 \rfloor}$, proving the claim. $\square$

Lemma 3.7 (Five-generator height-three cross-meet). *Let*

$$\mathcal{L} = \{\emptyset, D_1, D_2, D_3, D_4, D_5\}$$

be intersection-closed, with the D_i distinct and nonempty. Let $\mathcal{A}, \mathcal{B}$ be intersection-closed families of height at most three, both containing $\emptyset$. Suppose every member of either family is a union of a subcollection of the D_i and

$$A \cap B \in \mathcal{L} \quad (A \in \mathcal{A}, B \in \mathcal{B}).$$

Then

$$\min\{|\mathcal{A}|, |\mathcal{B}|\} \leq 11.$$

Proof. For every union X of the generators define

$$\sigma(X) = \{i \in [5] : D_i \subseteq X\}.$$

Then $X = \bigcup_{i \in \sigma(X)} D_i$, so σ is injective, preserves order, and preserves intersections. The assertion therefore reduces exactly, independently of the original ground size, to the following finite statement in B_5 : no two intersection-closed height-three subfamilies, each containing zero and having at least twelve members, have all their cross intersections in a six-element intersection-closed family consisting of zero and five distinct nonzero elements.

Here is the exhaustive certificate. In a nontrivial meet family of height at most three containing zero, the atoms are pairwise disjoint, all nonatoms form an antichain, and the meet of two distinct nonatoms is zero or an atom. Conversely, these conditions together with closure of each nonatom against

every atom construct every such family; the trivial family $\{\emptyset\}$ is handled separately. The primary enumeration finds 202 disjoint atom families, then 15,066 height-three intersection-closed families in total, of which 711 have at least twelve members. It also finds 12,656 admissible six-element meet families and tests all

$$\binom{712}{2} = 253,116$$

unordered pairs of large families. No pair is compatible. This search is slightly relaxed: it does not require the large families to lie in the union lattice of the selected five generators, so a negative result remains safe.

The independent implementation begins instead with all 7,580 antichains of $B_5 \setminus \{\emptyset\}$, combines them with the 202 possible atom families, directly checks every intersection, and independently recovers the same 711, 12,656, and 253,116 counts with no compatible pair. The two fail-closed exact programs and their deterministic outputs are described in Section 9. $\square$

Lemma 3.8 (Near-extremal bound). *Let $\mathcal{C}$ be a finite intersection-closed family containing $\emptyset$ and having height at most three. Suppose every member of $\mathcal{C}$ is a union of a subcollection of five fixed nonempty sets $G_1, \dots, G_5$. If $|\mathcal{C}| \geq 15$, then the G_i are pairwise disjoint, every G_i belongs to $\mathcal{C}$, and*

$$\mathcal{C} = \{\emptyset, G_1, \dots, G_5\} \cup \{G_i \cup G_j : \{i, j\} \in E\},$$

where $E \subseteq \binom{[5]}{2}$ has size nine or ten. In particular, $|\mathcal{C}| \in \{15, 16\}$, and a point of one generator belongs to at most five members of $\mathcal{C}$.

Proof. For $A \in \mathcal{C}$ put

$$\sigma(A) = \{i : G_i \subseteq A\}.$$

Every member is the union of all generators which it contains. Thus σ is an injective order embedding, preserves intersections, and identifies $\mathcal{C}$ with an intersection-closed height-three subfamily $\mathcal{S}$ of B_5 containing zero.

Let a be the number of atoms of $\mathcal{S}$. The atoms are pairwise disjoint. All remaining nonzero members are nonatoms and form an antichain: an atom below two comparable nonatoms would give a four-member chain. Thus there are at most ten nonatoms. We claim that $|\mathcal{S}| \geq 15$ forces $a = 5$. If $a \leq 4$ and there were ten nonatoms, equality in Sperner's theorem on B_5 would make them all two-subsets or all three-subsets. For completeness, the connected 3-regular incidence graph between the ten two-subsets and ten three-subsets shows that a mixed antichain has fewer than ten members: every nonempty proper set on one side has strictly larger neighbourhood. All two-subsets force all five singletons into $\mathcal{S}$ by intersection closure. All three-subsets force all two-subsets into $\mathcal{S}$, and these ten sets cannot be pairwise-disjoint atoms. Both alternatives contradict $a \leq 4$. Hence $a \leq 4$ gives at most nine nonatoms and at most fourteen members.

Five nonempty disjoint atoms in B_5 are the five singletons. Their preimages are exactly $G_1, \dots, G_5$, so the generators belong to $\mathcal{C}$ and are pairwise disjoint. There are at least nine nonatoms. Distinct nonatoms meet in zero

or a singleton, so no two contain the same two-subset of $[5]$. Assign to every nonatom all its internal two-subsets. These assigned collections are disjoint among the ten pairs. If one nonatom had size at least three, it would consume at least three pairs and the other eight would consume at least one each, requiring eleven pairs. Therefore all nonatoms are two-subsets, and there are nine or ten of them. The stated form and incidence bound follow. $\square$

Proof of Theorem 1.7. Theorem 1.6 leaves $q_0 + q_2 = 3$. Lemma 3.4 excludes $(0, 3)$.

Suppose first that $(q_0, q_2) = (1, 2)$. Relabel so that $q_{12} = q_{13} = 1$ and $q_{23} = 0$. Rows 2, 3 have unique possible private witnesses x_3, x_2 and attain their restricted matching cap five. Hence the unique support-zero member A is a lower endpoint in both rows. The union of $A \cup \{x_2\}$ and $A \cup \{x_3\}$ lies in the unique 23-fiber, whose member is $C_1 = R \cup \{x_2, x_3\}$. Therefore $A = R$.

The singleton complement-family sizes are

$$|\mathcal{C}_1| = r + u + 1, \quad |\mathcal{C}_2| = |\mathcal{C}_3| = r + u + 2,$$

while $|\mathcal{C}_{123}| = r + u + 1$ and $t = 2r + 2u + 6$. The unique 23-fiber and (12) give $R_2 \cap R_3 = \emptyset$. A residual coordinate outside $R_2 \cup R_3$ is omitted at most $(r + u + 1) + 2 + (r + u) = t - 3$ times; one in $R_2 \setminus R_1$ or $R_3 \setminus R_1$ is omitted at most $(r + u + 2) + 2 + (r + u) = t - 2$ times. Since every residual coordinate is omitted at least t times,

$$(13) \quad R = R_2 \dot{\cup} R_3, \quad R_1 = R.$$

Let $D_{12} \neq \emptyset$ be the complement of the extra 12-fiber member. Row-2 saturation gives $D_{12} \in \mathcal{C}_1$. By symmetry let row 1 witness x_2 . Then $D_{12} \notin \mathcal{C}_2$, or the associated singleton-2 member would be a third lower endpoint containing x_2 . Choose nonempty $E_2 \in \mathcal{C}_2$. For each $z \in E_2$, (13) supplies $E_1 \in \mathcal{C}_1$ containing z . Since $\mathcal{C}_{12} = \{\emptyset, D_{12}\}$, (12) gives $E_1 \cap E_2 = D_{12}$. Coordinatewise this proves $E_2 \subseteq D_{12}$, while the same equality gives $D_{12} \subseteq E_2$. Thus $E_2 = D_{12}$, a contradiction.

Next take $(q_0, q_2) = (2, 1)$ and relabel the exceptional pair as 12. Row 3 must match both support-zero members, and each other row matches at least one. If those members are A, B , choose a support-zero endpoint of row 1 and the other endpoint in row 3. Their matched extensions union through the unique 13-fiber, forcing $A \cup B = R$. The two-member support-zero fiber is union-closed, so it contains R ; a residual coordinate is omitted there at most once.

Here $|\mathcal{C}_3| = r + u + 1$, $|\mathcal{C}_{123}| = r + u + 1$, and $t = 2r + 2u + 4$. The unique 13- and 23-fibers make R_3 disjoint from $R_1 \cup R_2$. If $z \in R_3$, it is omitted at most $1 + (r + u + 1) + 1 + (r + u) = t - 1$ times, a contradiction. Hence $R_3 = \emptyset$, impossible because $\mathcal{C}_3$ has more than one member and contains $\emptyset$.

Finally let $(q_0, q_2) = (3, 0)$. The three pair fibers are unique, so R_1, R_2, R_3 are pairwise disjoint. Now

$$|C_i| = r + u - 1, \quad |C_{123}| = r + u + 1, \quad t = 2r + 2u + 2.$$

Any residual coordinate is omitted at most $3 + (r + u - 1) + (r + u) = t$ times. Equality is necessary for every coordinate, so all three support-zero members omit every coordinate of R . Having empty private support as well, all three would be the empty set, contradicting distinctness. This eliminates the last $q_0 + q_2 = 3$ type.

It remains to eliminate $q_0 + q_2 = 4$. We first record which private pairs can be nonfull. Here

$$s = 4 - 2q_0 + 2u, \quad t = 2r + 9 - 2q_0 + 2u,$$

so t is odd. For a private pair i, j , with remaining index k , the general fiber count in (10) gives

$$d_{ij} = r + 2 + u + q_{ij}, \quad \frac{t+1}{2} - d_{ij} = 3 - q_0 - q_{ij} = q_2 - 1 - q_{ij}.$$

If the pair were nonfull, the critical-pair dichotomy would require the last difference to be nonpositive. Equality is impossible by Lemma 3.3; hence $q_{ij} \geq q_2$. When $q_2 > 0$, the three pair weights sum to q_2 , so this can occur only when all pair mass is concentrated on the single edge ij . Thus every nonconcentrated positive- q_2 pattern is all-full, while in a concentrated pattern only its occupied edge requires a separate full/nonfull split. When $q_2 = 0$, this inequality does not restrict how many private pairs may be nonfull; the $(4, 0)$ proof below deliberately uses no pair-fullness assertion.

We first exclude the two cases with $q_0 \geq 3$.

First take $(q_0, q_2) = (4, 0)$. Every pair fiber consists only of its critical coatom, so its residual-complement family is $\{\emptyset\}$. Equation (12) therefore makes R_1, R_2, R_3 pairwise disjoint. Here

$$s = 2u - 4, \quad t = 2r + 2u + 1,$$

the three singleton complement families have size $r + u - 2$, and the full-support complement family has size $r + u + 1$.

Let M be the maximum of the four-member empty-support fiber. Its other members are $M \setminus D_1, M \setminus D_2, M \setminus D_3$ for pairwise disjoint nonempty blocks. A coordinate of M is therefore omitted in this fiber either once or zero times. By disjointness of the R_i , it is omitted in at most one singleton fiber, and the universal member reduces the full-support contribution by one. Its total number of omissions is at most

$$1 + (r + u - 2) + (r + u) = t - 2,$$

and an uncovered coordinate of M has the still smaller bound $t - 3$. Every residual coordinate must be omitted at least t times, so M has no coordinate. This makes M the empty set, contrary to the empty-set-free reduction.

It remains to exclude $(q_0, q_2) = (3, 1)$. Relabel so that $q_{12} = 1$ and put $K = r + u$. Then

$$(14) \quad |\mathcal{C}_1| = |\mathcal{C}_2| = K - 1, \quad |\mathcal{C}_3| = K, \quad |\mathcal{C}_{123}| = K + 1, \quad t = 2K + 3.$$

The unique 13- and 23-fibers give $R_3 \cap (R_1 \cup R_2) = \emptyset$. If D_{12} is the nonempty complement of the extra 12-fiber member, then

$$(15) \quad R_1 \cap R_2 \subseteq D_{12}.$$

Suppose first that the critical pair 12 is full. All three singleton complement families contain $\emptyset$. A coordinate in R_3 is omitted at most

$$3 + 1 + (K - 1) + K = t$$

times, where the terms come from empty support, the extra pair member, the non-top members of $\mathcal{C}_3$, and the non-universal full-support members. Equality is necessary and forces the coordinate into D_{12} . A coordinate lying in only one of R_1, R_2 is omitted at most

$$3 + 1 + (K - 2) + K = t - 1.$$

Together with (15), these facts imply $D_{12} = R$. If $R_1 \cap R_2$ is empty, then $R_1 = R_2 = \emptyset$ and both corresponding complement families have size one. Otherwise two of their members have nonempty intersection; that intersection lies in $\{\emptyset, R\}$, so both members are R . Intersecting arbitrary members with this copy of R confines both families to $\{\emptyset, R\}$. Either conclusion contradicts their size $K - 1 \geq 9$.

Finally suppose pair 12 is nonfull. Direct substitution in Tian's exact three-trace formula gives endpoint matching sizes $c_1 = c_2 = 3$, a residual critical coordinate z_0 , and the strict trace chain

$$(16) \quad S \subsetneq R \setminus \{z_0\} \subsetneq (R \setminus \{z_0\}) \cup \{x_3\}.$$

The two endpoint matchings put the extra residual complement

$$D_{12} = R \setminus (S \cup \{z_0\})$$

in both $\mathcal{C}_1$ and $\mathcal{C}_2$. The top trace in the singleton-3 fiber has complement $\{z_0\}$. Thus a coordinate $y \neq z_0$ in only one of R_1, R_2 , or in $R_3 \setminus D_{12}$, is omitted at most $t - 1$ times. Using (15), this gives $R \subseteq D_{12} \cup \{z_0\}$. The displayed formula for D_{12} then forces

$$S = \emptyset, \quad D_{12} = R \setminus \{z_0\}.$$

Since $z_0 \in R_3$, equation (12) makes every member of $\mathcal{C}_1 \cup \mathcal{C}_2$ a subset of D_{12} . Intersecting it with the copy of D_{12} in the opposite family shows that it is either $\emptyset$ or D_{12} . Again both families have size at most two, contrary to (14).

The face $(q_0, q_2) = (2, 2)$. Put $K = r + u$. Then $t = 2K + 5$ and $|\mathcal{C}_{123}| = K + 1$. Up to relabelling, the pair weights are $(2, 0, 0)$ or $(1, 1, 0)$.

First take $q_{12} = 2$. Then

$$(17) \quad |\mathcal{C}_1| = |\mathcal{C}_2| = K, \quad |\mathcal{C}_3| = K + 2.$$

Suppose pair 12 is nonfull. Its three complements are $\emptyset, D, E$, where the nonfull normal form gives a residual critical coordinate z_0 , with $z_0 \notin D$, $z_0 \in E$, and $D \in \mathcal{C}_1 \cap \mathcal{C}_2$. The other two pair fibers are unique, so $R_3 \cap (R_1 \cup R_2) = \emptyset$. The omission count at z_0 is

$$2 + (K + 2) + 1 + K = t.$$

It has no slack; in particular, only U contains z_0 in the full-support fiber. The nonfull normal form also places the singleton-3 complement $\{z_0\}$ in $\mathcal{C}_3$. Hence for $y \in R_3 \setminus \{z_0\}$ the singleton-3 contribution is at most $K + 1$, while $y \notin D$ because $D \in \mathcal{C}_1$ and $R_3 \cap R_1 = \emptyset$. At most the complement E can contribute in the pair fiber, so the total is at most $t - 1$. A coordinate in only one of R_1, R_2 is omitted at most $t - 2$ times, and a coordinate outside all three singleton supports is omitted at most $2 + 2 + K = K + 4 < t$ times. Therefore $R_3 = \{z_0\}$. If $y \in R_1 \cap R_2$, choose actual complements in $\mathcal{C}_1, \mathcal{C}_2$ containing y ; their nonempty intersection is D or E , so $y \in D \cup E$. Together with $z_0 \in E$ and the preceding strict bounds, this proves

$$R_3 = \{z_0\}, \quad D \cup E = R.$$

Intersection closure gives $D \cap E \in \{\emptyset, D\}$. If it equals D , then $E = R$; the pair member with complement E has empty residual trace and injects the $K + 2$ singleton-3 traces into the $K + 1$ full-support traces. If $D \cap E = \emptyset$, let S be the bottom residual trace in the nonfull normal form. Then

$$D = R \setminus (S \cup \{z_0\}), \quad E = S \cup \{z_0\},$$

where the second identity uses $D \cup E = R$ and $D \cap E = \emptyset$. Thus E , although introduced as a residual complement, is also the third-full-trace residual set. Its union with every singleton-3 trace contains z_0 and so, by the no-slack count, is U . Hence every such trace contains D . Union with the canonical z_0 -pair trace, whose residual set is D , again injects $K + 2$ traces into $K + 1$. Both alternatives are impossible.

Suppose instead pair 12 is full, and write $\mathcal{C}_{12} = \{\emptyset, D, E\}$. A coordinate in only one of R_1, R_2 is omitted at most $t - 2$ times. A coordinate in R_3 can reach t only by belonging to $D \cap E$. The unique 13- and 23-fibers make R_3 disjoint from $R_1 \cup R_2$, while a coordinate outside all three singleton supports is omitted at most $2 + 2 + K = K + 4 < t$ times. If $y \in R_1 \cap R_2$, choose actual complements in $\mathcal{C}_1, \mathcal{C}_2$ containing y . Their nonempty intersection is D or E , so $y \in D \cup E$. Hence

$$R = R_3 \dot{\cup} (R_1 \cap R_2), \quad R_3 \subseteq D \cap E, \quad D \cup E = R.$$

If $D \cap E$ is nonempty, intersection closure makes one of D, E the larger; it is then R and gives the same $K + 2$ into $K + 1$ injection. If the intersection is empty, $R_3 = \emptyset$ and D, E partition $R = R_1 \cap R_2$. For $A \in \mathcal{C}_1$ and $y \in A$, choose $B_y \in \mathcal{C}_2$ containing y . The nonempty intersection $A \cap B_y$ belongs to $\{D, E\}$, is contained in A , and contains y . Thus A is a union of a subcollection of D, E . The same argument applies to $\mathcal{C}_2$, so every member of either family is one of $\emptyset, D, E, R$, contradicting its size $K \geq 10$.

For split weights, relabel $q_{12} = q_{13} = 1$ and $q_{23} = 0$. All pairs are full and

$$|\mathcal{C}_1| = K, \quad |\mathcal{C}_2| = |\mathcal{C}_3| = K + 1.$$

Omission capacity and the unique 23-fiber give

$$R = (R_1 \cap R_2) \dot{\cup} (R_1 \cap R_3), \quad R_1 = R.$$

Writing $A = R_2$ and $B = R_3$, let D, E be the unique nonempty extra complements in the 12- and 13-fibers. For $a \in A$, choose complements in $\mathcal{C}_1, \mathcal{C}_2$ containing a . Their intersection is the unique nonempty 12 complement D , so $A \subseteq D$; since that intersection lies in a member of $\mathcal{C}_2$, also $D \subseteq R_2 = A$. Hence $D = A$, and similarly $E = B$. If either block is empty its singleton-complement family is trivial. Otherwise, for nonempty $F \in \mathcal{C}_2$ and $a \in F$, choose $G \in \mathcal{C}_1$ containing a . Then $F \cap G = D = A$, so $A \subseteq F \subseteq R_2 = A$. Thus $\mathcal{C}_2 \subseteq \{\emptyset, A\}$, a contradiction to its size $K + 1$. This excludes $(2, 2)$.

The face $(q_0, q_2) = (1, 3)$. Again put $K = r + u$. If $e_i = q_{ij} + q_{ik}$ is the pair degree, then

$$(18) \quad t = 2K + 7, \quad |\mathcal{C}_i| = K + 4 - e_i, \quad |\mathcal{C}_{123}| = K + 1.$$

When all pairs are full, a coordinate occurring in exactly one R_i is omitted at most

$$1 + (|\mathcal{C}_i| - 1) + 3 + K = t - e_i$$

times. Thus positive degree forbids one-support coordinates. A coordinate outside all three singleton supports is omitted at most $1 + 3 + K = K + 4 < t$ times, so no such coordinate exists.

For weights $(2, 1, 0)$, relabel the zero edge as 23 and the weight-one edge as 13. Every degree is positive, so $R = (R_1 \cap R_2) \dot{\cup} (R_1 \cap R_3)$ and $R_1 = R$. If the latter block $R_1 \cap R_3$ is empty, $\mathcal{C}_3$ is trivial. Otherwise, for an arbitrary coordinate y in that block choose complements in $\mathcal{C}_1, \mathcal{C}_3$ containing y . Their intersection is the unique nonempty 13 complement and is contained in R_3 . Since y was arbitrary, that fixed complement contains every coordinate of R_3 , so it equals the whole block. Repeating the same intersection with an arbitrary nonempty member of $\mathcal{C}_3$ confines $\mathcal{C}_3$ to the empty complement and that block, contradicting its size $K + 3$. For weights $(1, 1, 1)$, every coordinate lies in at least two supports. For $A \in \mathcal{C}_i$ and $y \in A$, choose a member of another singleton family containing y ; its nonempty intersection with A is one of the two incident pair complements and lies inside A . Thus A is a union of a subcollection of those two complements. This gives $|\mathcal{C}_i| \leq 4 < K + 2$.

It remains on this face to take concentrated weights $q_{12} = 3$. If pair 12 is full, then

$$R = R_3 \dot{\cup} (R_1 \cap R_2), \quad R_1 = R_2.$$

For $A \in \mathcal{C}_1$ and $y \in A$, choose $B_y \in \mathcal{C}_2$ containing y . Then $A \cap B_y$ is a nonempty 12 complement, lies in A , and contains y . Thus every member of $\mathcal{C}_1$ is a union of a subcollection of the three nonempty 12 complements. Hence $|\mathcal{C}_1| \leq 8 < K + 1$. If pair 12 is nonfull, write its complements as

$\emptyset, D, E, F$, where D avoids the residual witness z_0 , belongs to both endpoint singleton families, and E, F contain z_0 . The exact witness count

$$1 + (K + 4) + 2 + K = t$$

has no slack. The same nonfull normal form puts the singleton-3 complement $\{z_0\}$ in $\mathcal{C}_3$. Hence, for $y \in R_3 \setminus \{z_0\}$, at least that singleton-3 member does not omit y , and D cannot contain y because $R_3 \cap (R_1 \cup R_2) = \emptyset$. The resulting omission bound is at most $t - 1$. Together with the analogous strict bound for coordinates in only one of R_1, R_2 , this gives $R_3 = \{z_0\}$ and $R_1 = R_2 = R \setminus \{z_0\}$. Cross intersections avoid z_0 and are therefore only $\emptyset$ or D ; coordinatewise choice makes $D = R \setminus \{z_0\}$. Since D belongs to both families, each has at most two members, contrary to size $K + 1$. Thus (1, 3) is impossible.

The face $(q_0, q_2) = (0, 4)$. Now

$$(19) \quad t = 2K + 9, \quad |\mathcal{C}_i| = K + 6 - e_i, \quad |\mathcal{C}_{123}| = K + 1.$$

In the all-full case a coordinate occurring only in R_i is omitted at most $t - e_i$ times. A coordinate outside all three singleton supports is omitted at most $4 + K < t$ times. For nonconcentrated weights, consider first $(3, 1, 0)$ and $(2, 2, 0)$, with zero edge 23 and lighter edge 13 of weight $b = 1$ or 2. All degrees are positive. Hence no coordinate has exactly one singleton support, and the unique 23-fiber gives

$$R = (R_1 \cap R_2) \dot{\cup} (R_1 \cap R_3).$$

For $A \in \mathcal{C}_3$ and $y \in A$, choose $B_y \in \mathcal{C}_1$ containing y . The nonempty intersection $A \cap B_y$ is one of the b nonempty 13 complements, is contained in A , and contains y . Thus every member of $\mathcal{C}_3$ is a union of a subcollection of those b complements, so $|\mathcal{C}_3| \leq 2^b$, contradicting respectively $K + 5 > 2$ and $K + 4 > 4$. For weights $(2, 1, 1)$, the degree vector is $(3, 3, 2)$ and every coordinate lies in at least two singleton supports. For $A \in \mathcal{C}_i$ and $y \in A$, choose another singleton-complement family through y . Their nonempty intersection is an incident pair complement contained in A . These incident complements therefore cover A , so every member of $\mathcal{C}_i$ is a union of a subcollection of its e_i incident pair complements. The bounds $|\mathcal{C}_i| \leq 2^{e_i}$ contradict the exact sizes $(K + 3, K + 3, K + 4)$.

Finally take concentrated weights $q_{12} = 4$. If pair 12 is nonfull, its complements are $\emptyset, D, E_1, E_2, E_3$, where D avoids and the E_j contain a residual witness z_0 , and $D \in \mathcal{C}_1 \cap \mathcal{C}_2$. The count

$$(K + 6) + 3 + K = t$$

has no slack. The nonfull normal form includes $\{z_0\} \in \mathcal{C}_3$. Hence for $y \in R_3 \setminus \{z_0\}$ the singleton-3 contribution drops by one; also D cannot contain y because the unique 13- and 23-fibers make R_3 disjoint from $R_1 \cup R_2$. Thus y is omitted at most $t - 1$ times. The analogous strict bound for a coordinate in only one of R_1, R_2 gives $R_3 = \{z_0\}$ and $R_1 = R_2 = R \setminus \{z_0\}$. Cross intersections avoid z_0 , so they are only $\emptyset$ or D ; hence $D = R \setminus \{z_0\}$ and

both endpoint singleton families have at most two members, contradicting their size $K + 2$.

If pair 12 is full, omission capacity gives

$$R = R_3 \dot{\cup} S, \quad R_1 = R_2 = S.$$

Put $\mathcal{L} = \mathcal{C}_{12}$; it consists of the empty set and four distinct nonempty complements and is intersection-closed. The full 13- and 23-pairs put the empty complement in $\mathcal{C}_2$ and $\mathcal{C}_1$, respectively. For $A \in \mathcal{C}_1$ and $y \in A \subseteq S = R_2$, choose $B_y \in \mathcal{C}_2$ containing y . Then $A \cap B_y$ is a nonempty member of $\mathcal{L}$, is contained in A , and contains y ; hence A is a union of a subcollection of the four nonempty members of $\mathcal{L}$. The same holds with the two singleton families interchanged, and (12) puts every cross intersection in $\mathcal{L}$. All hypotheses of Lemma 3.5 are now verified, so it gives

$$\min\{|\mathcal{C}_1|, |\mathcal{C}_2|\} \leq 8,$$

contrary to their common size $K + 2 \geq 12$.

All five faces of $q_0 + q_2 = 4$ are excluded. It remains to exclude the six faces of $q_0 + q_2 = 5$. Put $K = r + u$ and $e_i = q_{ij} + q_{ik}$. The balance equations give

$$(20) \quad s = 5 - 2q_0 + 2u, \quad t = 2K + 10 - 2q_0, \quad |\mathcal{C}_i| = K + 7 - 2q_0 - e_i.$$

The full-support complement family has $K + 1$ members and contains $\emptyset$, so at most K of its members omit a fixed residual coordinate.

First take $(q_0, q_2) = (5, 0)$. Every pair fiber is unique, hence the three R_i are pairwise disjoint, and each singleton complement family has size $K - 3$. Let M be the maximum of the five-member support-zero fiber. Its four proper members have complements in M which are pairwise disjoint and nonempty. A coordinate of M is therefore omitted at most once in this fiber, in at most one singleton fiber, and at most K times in full support. The total is at most

$$1 + (K - 3) + K = 2K - 2 < t = 2K.$$

Thus M has no coordinate, contrary to the empty-set-free reduction.

Next let $(q_0, q_2) = (4, 1)$ and relabel the extra pair member onto support 12. Let M now denote the maximum of the four-member support-zero fiber. Write

$$\mathcal{C}_{12} = \{\emptyset, D\}, \quad \mathcal{C}_{13} = \mathcal{C}_{23} = \{\emptyset\}.$$

The zero edges give $R_3 \cap (R_1 \cup R_2) = \emptyset$ and $R_1 \cap R_2 \subseteq D$. Their private critical pairs are at the exact threshold and hence are full, so $\mathcal{C}_1, \mathcal{C}_2$ contain $\emptyset$. Here

$$|\mathcal{C}_1| = |\mathcal{C}_2| = K - 2, \quad |\mathcal{C}_3| = K - 1, \quad t = 2K + 2.$$

If $z \in M$ lies in neither singleton support, only in one of R_1, R_2 , or in R_3 , its omission count is at most, respectively,

$$1 + 1 + K, \quad 1 + (K - 3) + 1 + K, \quad 1 + (K - 1) + 1 + K,$$

each strictly below t . Hence $M \subseteq R_1 \cap R_2 \subseteq D$. Intersecting D first with $R \setminus M$ and then with $R \setminus (M \setminus D_i)$, where the three D_i are the disjoint nonempty complement blocks of the proper support-zero members, gives

$$D = M, \quad D_i \in \{\emptyset, M\}.$$

This is impossible for three pairwise disjoint nonempty blocks.

For $(q_0, q_2) = (3, 2)$, either pair-mass partition has a source i of degree $e_i = 2$ whose opposite edge has weight zero. That critical pair is full, so $\emptyset \in \mathcal{C}_i$. A coordinate occurring only in R_i is omitted at most

$$3 + (|\mathcal{C}_i| - 1) + 2 + K = t - 1$$

times. Hence $R_i \subseteq R_j \cup R_k$. Pointwise cross-intersection expresses every member of $\mathcal{C}_i$ as a union of the two incident nonempty pair complements. Therefore $|\mathcal{C}_i| \leq 4$, contrary to $|\mathcal{C}_i| = K - 1 \geq 9$.

For $(q_0, q_2) = (2, 3)$, choose a source of degree $e_i \geq 2$. Its opposite edge has weight $3 - e_i \leq 1$ and is strictly below the nonfull threshold. As above, the only- R_i omission bound is

$$2 + (|\mathcal{C}_i| - 1) + 3 + K = t + 1 - e_i < t,$$

and $\mathcal{C}_i$ is generated by its e_i incident pair complements. Here $e_i \in \{2, 3\}$. The resulting bounds $|\mathcal{C}_i| \leq 2^{e_i}$ contradict, respectively, $K + 1 \geq 11$ and $K \geq 10$.

For $(q_0, q_2) = (1, 4)$, every nonconcentrated pair-mass partition has a source of degree $e_i \in \{2, 3\}$ whose opposite edge has weight $4 - e_i \in \{2, 1\}$ and is full. The same argument gives $|\mathcal{C}_i| \leq 2^{e_i}$, whereas the exact sizes are $K + 3 \geq 13$ and $K + 2 \geq 12$. In the remaining concentrated pattern $q_{12} = 4$, the zero edges are full and unique. They make R_3 disjoint from $R_1 \cup R_2$, while the strict degree-four omission bounds give $R_1 = R_2$. The two endpoint complement families, each of size $K + 1$, contain $\emptyset$, are generated by the four nonempty 12-complements, and have every cross intersection in $\mathcal{C}_{12}$. Lemma 3.5 gives balanced size at most eight, a contradiction.

It remains to take $(q_0, q_2) = (0, 5)$. The five pair-mass partitions are

$$(5, 0, 0), (4, 1, 0), (3, 2, 0), (3, 1, 1), (2, 2, 1).$$

In every nonconcentrated partition all private pairs are full: indeed,

$$\left\lceil \frac{t+1}{2} \right\rceil - d_{ij} = 4 - q_{ij},$$

and weights below four are strictly below the nonfull threshold, while weight four is full by Lemma 3.3. Such a partition has a source of degree $e_i \in \{2, 3, 4\}$. The only- R_i omission bound is $t + 1 - e_i < t$, so $\mathcal{C}_i$ is generated by its e_i incident pair complements. Since all three private pairs are full, $\emptyset \in \mathcal{C}_i$.

We now justify explicitly the height hypothesis used at this point. A strict four-member chain in a singleton fiber could be extended by successive unions with the two missing private-cover members; the private supports

increase strictly at both steps, producing a six-member chain in the original family. Height five therefore makes every singleton fiber, and hence its order-reversing residual-complement family, have height at most three. Lemma 3.6 applies. For $e_i = 2, 3, 4$ its bounds are 5, 7, 11, whereas (20) gives at least $K + 5, K + 4, K + 3$, respectively, and these are at least 15, 14, 13. Thus every nonconcentrated partition is impossible.

Finally relabel the concentrated partition as $q_{12} = 5$ and $q_{13} = q_{23} = 0$. The two zero pairs are full and unique, so $R_3 \cap (R_1 \cup R_2) = \emptyset$. The strict degree-five omission bounds give $R_1 = R_2$. Put

$$\mathcal{L} = \mathcal{C}_{12} = \{\emptyset, D_1, D_2, D_3, D_4, D_5\}.$$

For $A \in \mathcal{C}_1$ and $y \in A$, choose $B_y \in \mathcal{C}_2$ containing y . Then $A \cap B_y$ is a nonempty member of $\mathcal{L}$, is contained in A , and contains y . Hence $\mathcal{C}_1$ is generated by the five nonempty members of $\mathcal{L}$, and symmetrically so is $\mathcal{C}_2$. The opposite zero pairs put $\emptyset$ in both endpoint families; the singleton-fiber chain-extension argument just given makes both heights at most three; and every cross intersection lies in $\mathcal{L}$ by (12). Lemma 3.7 gives

$$\min\{|\mathcal{C}_1|, |\mathcal{C}_2|\} \leq 11,$$

contrary to their common size $K + 2 \geq 12$. This argument never assumes that the weight-five pair is full. All six faces of $q_0 + q_2 = 5$ are excluded, so $q_0 + q_2 \geq 6$. $\square$

4. THE BOUNDARIES SIX, SEVEN, AND EIGHT

We complete the proof of Theorem 1.8. Put $h = q_0 + q_2$, $K = r + u$, and $e_i = q_{ij} + q_{ik}$. The balance identities are

$$(21) \quad t = 2K + h + 5 - 2q_0, \quad |\mathcal{C}_i| = K + h + 2 - 2q_0 - e_i, \quad |\mathcal{C}_{123}| = K + 1.$$

Every numerical gap in the boundary ledgers below is obtained by literal substitution of the displayed degree and fiber sizes into (21) and the corresponding displayed omission bound. Thus the ledgers and their arithmetic are part of the human proof; the exact programs described in Section 9 are regression audits, not substitutes for these deductions. Whenever the opposite pair jk is full, a point which occurs only in $R_i = \bigcup \mathcal{C}_i$ is omitted at most

$$(22) \quad q_0 + (|\mathcal{C}_i| - 1) + q_2 + K = t + h - 4 - e_i$$

times. If this bound is strict, then $R_i \subseteq R_j \cup R_k$. Pointwise cross intersection consequently generates every member of $\mathcal{C}_i$ by its e_i incident nonempty pair complements. Hence

$$(23) \quad |\mathcal{C}_i| \leq H(e_i) := 1 + e_i + \binom{e_i}{\lfloor e_i/2 \rfloor}$$

by Lemma 3.6.

Lemma 4.1 (Strict and exact source reduction). *On boundary $h \in \{6, 7, 8\}$, suppose the opposite pair jk is full. If $e_i > h - 4$, then $R_i \subseteq R_j \cup R_k$ and (23) holds. If $e_i = h - 4$, equality in (22) forces the source-only point into every support-zero complement, every nonempty member of $\mathcal{C}_i$, every nonempty pair complement, and every nonempty full-support complement. On boundary eight this equality pattern also forces $R_i \subseteq R_j \cup R_k$.*

Proof. The strict assertion is (22). If $A \in \mathcal{C}_i$ and $y \in A$, a second singleton support through y supplies a member whose intersection with A is a nonempty incident pair complement contained in A . Taking the union pointwise proves generation. Equality in the four summands of (22) is termwise equality, proving the saturation assertion.

For $h = 8, e_i = 4$, suppose a source-only point z exists. Saturation makes z belong to every nonempty pair complement, every nonempty full-support complement, and every nonempty member of $\mathcal{C}_i$, but to no member of $\mathcal{C}_j \cup \mathcal{C}_k$. Any nonempty cross intersection involving source i , or involving sources j, k , would be a pair complement omitting z , contradicting saturation. Hence the three singleton support unions are pairwise disjoint. If $y \in R_j$, intersect a member of $\mathcal{C}_j$ through y with any pair or full-support complement through y ; the resulting nonempty complement would again omit z , so no such pair or full-support complement exists. Thus

$$o(y) \leq q_0 + |\mathcal{C}_j| \leq K + 10 - q_0 < t,$$

because on boundary eight $q_0 \leq 8$, so $t - (K + 10 - q_0) = K + 3 - q_0 > 0$. This forces $R_j = \emptyset$, although (21) gives $|\mathcal{C}_j| > 1$, a contradiction. This uses equality saturation and cross intersection only, not reverse localization for a full pair. $\square$

Lemma 4.2 (High-density meet family). *Let $\mathcal{A}$ be a finite intersection-closed family of distinct subsets of $U = \bigcup \mathcal{A}$. If every point of U belongs to at least $|\mathcal{A}| - 1$ members of $\mathcal{A}$, then $|\mathcal{A}| \leq 2$.*

Proof. The complement family $\mathcal{D} = \{U \setminus A : A \in \mathcal{A}\}$ is union-closed, and every point belongs to at most one member of $\mathcal{D}$. Two distinct nonempty members of $\mathcal{D}$ must therefore be disjoint. Their union is a third member, strictly larger than either, and reuses every point of both, a contradiction. Thus $\mathcal{D}$ has at most one nonempty member, in addition to a possible empty member, and $|\mathcal{A}| = |\mathcal{D}| \leq 2$. $\square$

Lemma 4.3 (Effective-generator leaf collapse). *Assume $R_c \subseteq R_a \cup R_b$ and that $\mathcal{C}_c$ is generated by its e_c incident nonempty pair complements. If $|\mathcal{C}_c| > H(m - 1)$, then at least m of those complements are contained in R_c . Consequently a coordinate outside R_c belongs to at most $e_c - m$ of them. If a leaf support R_a is thereby forced into R_c , then every member of $\mathcal{C}_a$ is generated by the q_{ac} complements on their common edge, and*

$$|\mathcal{C}_a| \leq 2^{q_{ac}}.$$

The same conclusions hold with two central endpoint families: a complement which is effective at either endpoint lies in their union, while at most the sum of the two inactive budgets can meet a point outside that union.

Proof. If fewer than m generators were effective, the height-three generator bound would give at most $H(m - 1)$ members. The pointwise assertion is then immediate. Once $R_a \subseteq R_c$, intersect each member of $\mathcal{C}_a$ pointwise with a member of $\mathcal{C}_c$ through the same coordinate. The resulting nonempty cross intersections are precisely the fixed ac complements and generate the original member by their union. The last assertion is the union bound for q_{ac} fixed sets. The two-endpoint form follows by taking the union of their inactive lists. $\square$

We record explicitly the projection mechanism used below. Let M be the maximum of the support-zero residual fiber and write its proper members as $M \setminus Q_\ell$. In residual-complement coordinates their complements are

$$P_0 = R \setminus M, \quad P_\ell = (R \setminus M) \cup Q_\ell.$$

If E belongs to any fixed private-support complement family, union closure applied to the corresponding original member and to M or $M \setminus Q_\ell$ stays in that same private-support fiber. Passing back to complements gives $E \cap P_0$ and $E \cap P_\ell$ in the same complement family. Hence every later use of a projection lemma has the required projection closure from these support-zero traces; pair fullness alone is never used as a localization principle.

Lemma 4.4 (Deletion projection). *Let M have pairwise-disjoint nonempty deletion blocks $Q_1, \dots, Q_b$, and put $P_0 = R \setminus M$, $P_\ell = P_0 \cup Q_\ell$. Suppose an intersection-closed family $\mathcal{L}$ is stable under intersection with every P_ℓ .*

- (a) *If $\mathcal{L} = \{\emptyset, D, E\}$ and $M \subseteq D \cup E$, then $b \leq 2$.*
- (b) *If $\mathcal{L} = \{\emptyset, A, B, C\}$, the three nonzero members have a common point in P_0 , and $M \subseteq A \cup B \cup C$, then $b \leq 2$.*

Proof. In (a), D, E are disjoint or nested. In the disjoint case their portions in M are two indivisible projection atoms; in the nested case the only possible nonempty difference atom is $E \setminus D$ (or symmetrically). Every Q_ℓ , being recovered by projection with P_ℓ , is a union of whole atoms, so three pairwise-disjoint nonempty blocks are impossible.

For (b), the common outside point makes every pairwise meet nonempty. Intersection closure therefore gives a unique minimum, say $A \subseteq B, C$, fixed by projection to P_0 . If B, C are incomparable, the only M -atoms are $B \setminus A$ and $C \setminus A$. If they are nested, the only possible atoms are two successive differences. Projection by each P_ℓ again makes every deletion block a union of whole atoms, so $b \leq 2$. $\square$

Lemma 4.5 (Pair-family projection capacity). *Let M , $Q_1, \dots, Q_b$, $P_0 = R \setminus M$, and $P_\ell = P_0 \cup Q_\ell$ be as in Lemma 4.4. Assume that every family below is intersection-closed and is mapped into itself by intersection with each P_ℓ , $0 \leq \ell \leq b$.*

- (a) If $\mathcal{L} = \{\emptyset, H\}$, then either $H \cap M = \emptyset$ or H meets at most one deletion block.
- (b) If $\mathcal{L} = \{\emptyset, D, E\}$, then $D \cup E$ meets at most two deletion blocks. If $D \cap E \cap P_0 \neq \emptyset$, then it meets at most one deletion block.

These conclusions apply to a private pair-complement family only after the projection-closure hypothesis has been supplied by the support-zero maximum and its proper traces; pair fullness by itself is not used as a localization principle.

Proof. For (a), $H \cap P_0$ is either $\emptyset$ or H . In the latter case $H \cap M = \emptyset$. In the former case, each $H \cap P_\ell = H \cap Q_\ell$ is either $\emptyset$ or H , and the disjointness of the Q_ℓ permits only one nonempty value.

For (b), intersection closure makes D, E disjoint or nested. In the disjoint case, part (a), applied separately to D and E , gives at most two occupied blocks. Suppose $D \subsetneq E$. The value $E \cap P_0$ is $\emptyset, D$, or E . In the last case neither member has an M -part; in the middle case only the single difference $E \setminus D$ can occupy a deletion block; in the first case the two nested atoms D and $E \setminus D$ occupy at most two blocks. If D and E share a point of P_0 , they cannot be disjoint, the smaller member is fixed by P_0 , and only the difference atom can meet M . This proves the sharper bound. $\square$

Lemma 4.6 (Deletion-block injection). *If the support-zero fiber is nonempty, then*

$$q_0 - 1 \leq q_2.$$

More precisely, its $q_0 - 1$ deletion blocks inject into the labelled nonempty complements in the three private pair fibers.

Proof. Let M be the support-zero maximum and $M \setminus Q_1, \dots, M \setminus Q_{q_0-1}$ its proper members. The Q_ℓ are pairwise-disjoint and nonempty. We first show that every $y \in M$ lies in a nonempty private-pair complement. Otherwise cross intersection prevents y from lying in two singleton supports. If it lies only in R_i , its support-zero, singleton, pair, and full-support omission contributions are bounded by

$$1, \quad |\mathcal{C}_i|, \quad 0, \quad K.$$

Using (21)—or its identical trace identity before a boundary is fixed—their sum is $t - (2 + e_i) < t$; with no singleton support it is smaller. This contradicts residual capacity.

Choose $y_\ell \in Q_\ell$ and, among all pair complements containing it, let G_ℓ be the intersection-minimum one in its labelled pair family. Projection against the proper support-zero trace gives

$$G_\ell \cap ((R \setminus M) \cup Q_\ell) \in \mathcal{C}_{ij}.$$

This set contains y_ℓ and is contained in G_ℓ ; minimality makes it equal to G_ℓ . Hence $\emptyset \neq G_\ell \cap M \subseteq Q_\ell$. The same labelled pair complement cannot be assigned to two disjoint deletion blocks. Choosing the first admissible

labelled complement therefore gives an injection into a set of cardinality q_2 . $\square$

4.1. Boundary six. Here the strict source threshold is $e_i \geq 3$. The partitions left after (23), with their terminal mechanisms, are

(q_0, q_2)	pair masses	applied lemma or terminal mechanism
(0, 6)	(4, 2, 0), (3, 3, 0)	Lemma 4.1 and $H(e)$
	(3, 2, 1), (2, 2, 2)	Lemma 4.1 and $H(e)$
	(6, 0, 0), (5, 1, 0), (4, 1, 1)	effective zero-edge collapse
(1, 5)	all four nonconcentrated partitions	Lemma 4.1 and $H(e)$
	(5, 0, 0)	five-generator endpoint capacity
(2, 4)	(3, 1, 0), (2, 1, 1)	Lemma 4.1
	(4, 0, 0), (2, 2, 0)	four-generator cross meet; leaf collapse
(3, 3)	(3, 0, 0), (2, 1, 0)	Lemma 4.1
	(1, 1, 1)	exact saturation and two-generator collapse
(4, 2)	(2, 0, 0), (1, 1, 0)	three deletion blocks, at most two atoms
(5, 1), (6, 0)	all	support-zero maximum projection

We first make the source-degree rows completely explicit. In each entry of the following table, choose a source with displayed degree e . Its opposite pair lies below the nonfull threshold (equality is handled by Lemma 3.3), so Lemma 4.1 applies. The final column compares the exact source size at the smallest $K = 10$ with $H(e)$; larger K only strengthens the contradiction.

(q_0, q_2)	pair masses	chosen e	$ \mathcal{C}_i _{K=10} > H(e)$
(0, 6)	(4, 2, 0)	4	$14 > 11$
	(3, 3, 0)	3	$15 > 7$
	(3, 2, 1)	3	$15 > 7$
	(2, 2, 2)	4	$14 > 11$
(1, 5)	(4, 1, 0)	4	$12 > 11$
	(3, 2, 0)	3	$13 > 7$
	(3, 1, 1)	4	$12 > 11$
	(2, 2, 1)	3	$13 > 7$
(2, 4)	(3, 1, 0)	3	$11 > 7$
	(2, 1, 1)	3	$11 > 7$
(3, 3)	(3, 0, 0)	3	$9 > 7$
	(2, 1, 0)	3	$9 > 7$

This proves exactly the source-reduction rows of the preceding ledger, not merely a numerical sample.

We next close the three rows left on $(q_0, q_2) = (0, 6)$. In $(5, 1, 0)$ the degree-five leaf family has size $K + 3 \geq 13$, so all five heavy-edge generators are effective and its support lies in the degree-six source. A point of the remaining leaf outside that source has at most the unique light-edge pair hit, giving

$$o(y) \leq (K + 7) + 1 + K = 2K + 8 < t = 2K + 11.$$

The leaf is consequently one-generated and has at most two members. In $(4, 1, 1)$ the two degree-five source families have all five generators effective. A coordinate outside both source supports has no pair hit and is omitted at most $(K + 6) + K < t$ times; the remaining leaf is generated by the two light-edge complements and has at most four members. In the concentrated row $(6, 0, 0)$, the zero pairs separate the isolated leaf and strict degree-six inclusion makes the endpoint supports equal. Each endpoint family has size $K + 2 \geq 12$, so at least five of the six generators are effective at each endpoint and at most one is inactive at both. An isolated-leaf coordinate is therefore omitted at most

$$(K + 8) + 1 + K = 2K + 9 < t,$$

forcing an empty leaf support although $|\mathcal{C}_3| = K + 8$.

For the concentrated $(q_0, q_2) = (1, 5)$ row, the five-generator cross-meet bound first gives $K + 1 \leq 11$, hence $K = 10$. The endpoint families then have size eleven and the isolated leaf family size sixteen. At least four of the five occupied-edge generators are effective at each endpoint, so at most one is ineffective at both. An isolated-leaf coordinate is omitted at most

$$1 + 16 + 1 + 10 = 28 < t = 29,$$

again forcing an empty support. This closes the entire $q_0 = 1$ face.

On $(q_0, q_2) = (2, 4)$, the concentrated row $(4, 0, 0)$ has equal endpoint supports and two endpoint families of size K . They are generated by the four occupied-edge complements, contain $\emptyset$, and have all cross intersections in that five-member pair family; the four-generator cross-meet bound gives size at most eight, contrary to $K \geq 10$. In the remaining split row $(2, 2, 0)$, all three pairs are full by Lemma 3.3. The degree-four source is generated by its four incident complements and has size $K \geq 10$, so all four are effective. The opposite zero edge separates the leaves. A leaf coordinate outside the source has

$$o(y) \leq 2 + (K + 1) + K = 2K + 3 < t = 2K + 7,$$

so both leaves lie in the source. Either leaf family is then generated by its two occupied-edge complements and has at most four members, not $K + 2 \geq 12$.

On $(q_0, q_2) = (3, 3)$ only $(1, 1, 1)$ remains. Every pair is full by Lemma 3.3 and $\mathcal{C}_{ij} = \{\emptyset, D_{ij}\}$. If a coordinate lies in only one singleton support, the exact bound

$$3 + (K - 1) + 3 + K = 2K + 5 = t$$

puts it outside the support-zero maximum M and in all three D_{ij} . Projection by $R \setminus M$ then makes all D_{ij} disjoint from M . But a point of M with at most one singleton support has bound $1 + (K - 1) + 3 + K < t$; hence it has two supports and belongs by cross intersection to some D_{ij} , a contradiction. Thus every residual coordinate has two singleton supports, and pointwise cross intersection generates each $\mathcal{C}_i$ by its two incident complements. This gives $|\mathcal{C}_i| \leq 4 < K$.

It remains to justify the projection faces. For $(q_0, q_2) = (4, 2)$ the support-zero maximum has three deletion blocks. In the concentrated partition $(2, 0, 0)$, zero edges separate the isolated support. A point of M occurring at only one endpoint has bound $2K + 1 < t = 2K + 3$; the only exact isolated-support case forces membership in both occupied-edge complements. Consequently $M \subseteq D \cup E$, where $\mathcal{C}_{12} = \{\emptyset, D, E\}$. Lemma 4.4(a) forbids three blocks. In the split partition $(1, 1, 0)$, every point of M must have two singleton supports, so $M \subseteq D \cup E$ for the two unique occupied-edge complements; Lemma 4.5(a) again permits at most two occupied blocks.

For $(q_0, q_2) = (5, 1)$, M has four deletion blocks and the unique pair complement D . The exact singleton sizes show that a point of M outside $R_1 \cap R_2$ is omitted at most $t - 1$; hence $M \subseteq D$. Projection by P_0 gives $D = M$, while projection by each P_ℓ makes the four nonempty deletion blocks equal to M , impossible. Finally, for $(q_0, q_2) = (6, 0)$ the singleton supports are pairwise disjoint and a point of M is omitted at most

$$1 + (K - 4) + K = 2K - 3 < t = 2K - 1.$$

Thus $M = \emptyset$, contradicting the empty-set-free reduction. Every partition on boundary six has now been covered, so the boundary is empty.

4.2. Boundary seven. The strict source threshold is $e_i \geq 4$. The complete unordered face ledger is

q_0	pair masses	mechanism
0	(7, 0, 0) (4, 3, 0), (3, 3, 1), (3, 2, 2) (6, 1, 0), (5, 2, 0) (5, 1, 1), (4, 2, 1)	concentrated endpoint strict degree-four source effective leaf collapse effective leaf collapse
1	(4, 2, 0), (3, 2, 1), (2, 2, 2) (5, 1, 0), (4, 1, 1), (3, 3, 0) (6, 0, 0)	strict degree-four source effective leaf collapse equality collapse at $K = 10$
2	(5, 0, 0), (4, 1, 0), (3, 2, 0) (3, 1, 1), (2, 2, 1)	equality or effective leaf collapse equality or effective leaf collapse
3	(4, 0, 0), (3, 1, 0), (2, 2, 0), (2, 1, 1)	source generation and equality collapse
4	(3, 0, 0), (2, 1, 0), (1, 1, 1)	three-block projection
5	(2, 0, 0), (1, 1, 0)	four-block projection
6	(1, 0, 0)	five-block projection
7	(0, 0, 0)	zero-pair omission deficit

For $q_0 \leq 3$, every nonconcentrated partition either has a strict source whose family is too large for its effective incident generators, or has a zero edge. A zero edge separates the two leaf supports; effectiveness at the central source puts every pair complement in the central support, and the remaining leaf is generated by at most two sets. We now give the concentrated calculation in full, since its equality endpoints require a separate saturation argument.

Relabel so that

$$q_{12} = q_2 = 7 - q_0, \quad q_{13} = q_{23} = 0, \quad 0 \leq q_0 \leq 3.$$

The two zero pairs are full by the critical-pair dichotomy together with Lemma 3.3. The common trace identities are

$$(24) \quad t = 2K + 12 - 2q_0, \quad |\mathcal{C}_1| = |\mathcal{C}_2| = K + 2 - q_0, \quad |\mathcal{C}_3| = K + 9 - 2q_0.$$

Since the endpoint degree is at least four, strict source reduction gives

$$R_1 = R_2 =: S, \quad S \cap R_3 = \emptyset.$$

Both endpoint families are generated by the same q_2 nonempty complements on edge 12. If each endpoint has at least m effective generators, then a complement meeting R_3 is ineffective at both endpoints, so a point of R_3 lies in at most $q_2 - m$ pair complements. We use $H(3) = 7$ and $H(4) = 11$ from Lemma 3.6.

If $q_0 = 0$, the endpoint size is $K + 2 \geq 12$, so at least five of the seven generators are effective and at most two can meet R_3 . For $z \in R_3$,

$$o(z) \leq (K + 9) + 2 + K = 2K + 11 < t = 2K + 12,$$

a contradiction. If $q_0 = 1$ and $K \geq 11$, the endpoint size is at least twelve, so five of the six generators are effective and

$$o(z) \leq 1 + (K + 7) + 1 + K = 2K + 9 < t = 2K + 10.$$

Thus only $K = 10$ remains. There the endpoint size is eleven, hence at least four generators are effective and at most two meet R_3 ; consequently

$$o(z) \leq 1 + (K + 7) + 2 + K = t.$$

Residual capacity forces equality term by term. In particular every member of $\mathcal{C}_3$ contains z . This holds for every $z \in R_3$, so all $K + 7 = 17$ distinct members of $\mathcal{C}_3$ equal R_3 , impossible. If $R_3 = \emptyset$, the same conclusion that there is at most one distinct member is immediate.

If $q_0 = 2$ and $K \geq 12$, the endpoint size K forces all five generators to be effective, and

$$o(z) \leq 2 + (K + 5) + K = 2K + 7 < t = 2K + 8.$$

Thus $K \in \{10, 11\}$. At either residual value at least four generators are effective, at most one meets R_3 , and

$$o(z) \leq 2 + (K + 5) + 1 + K = t.$$

Termwise equality again makes every member of $\mathcal{C}_3$ equal to R_3 , contradicting $|\mathcal{C}_3| = K + 5 > 1$.

Finally, for $q_0 = 3$ the endpoint size is $K - 1 \geq 9 > H(3)$, so all four generators are effective. No occupied-edge complement meets R_3 , and

$$o(z) \leq 3 + (K + 3) + K = 2K + 6 = t.$$

The same equality collapse contradicts $|\mathcal{C}_3| = K + 3 > 1$ for every $K \geq 10$. This completes every concentrated row without restricting the ground set.

For reference, the terminal strict calculations for the nonconcentrated rows not removed immediately by Lemma 4.1 are shown below. The “gap” is t minus the displayed safe omission bound for a leaf point outside the central support; the last column is the number of fixed generators remaining after the resulting support inclusion.

q_0	pair masses	minimum gap	leaf generators
0	$(6, 1, 0), (5, 2, 0), (5, 1, 1), (4, 2, 1)$	2, 3, 2, 5	1, 2, 2, 3
1	$(5, 1, 0), (4, 1, 1), (3, 3, 0)$	2, 4, 3	1, 2, 3
2	$(4, 1, 0), (3, 2, 0), (3, 1, 1), (2, 2, 1)$	1, 2, 3, 3	1, 3, 2, 2
3	$(3, 1, 0), (2, 2, 0), (2, 1, 1)$	1, 2, 1	3, 2, 2

After inclusion, Lemma 4.3 bounds the corresponding leaf family by 2, 4, 8 according as the last entry is 1, 2, 3; the exact sizes from (21) are at least $K \geq 10$. This verifies every nonconcentrated row in the ledger without an implicit finite-ground argument.

We spell out the projection endpoint $q_0 = 4$, since it is the first place where a mere generator count is insufficient. Put $q_2 = 3$ and $t = 2K + 4$; then $|C_i| = K + 1 - e_i$. Let M be the support-zero maximum and write its three proper members as $M \setminus Q_\ell$, where Q_1, Q_2, Q_3 are pairwise-disjoint and nonempty.

For the concentrated partition $(3, 0, 0)$, the zero edges separate R_3 from $R_1 \cup R_2$. Each endpoint family has size $K - 2 \geq 8$. If, say, $R_1 \subseteq R_2 \cup R_3$, it

would be generated by the three occupied-edge complements and have size at most seven. Thus a source-1-only coordinate z exists. Its safe bound

$$4 + (K - 3) + 3 + K = 2K + 4 = t$$

is exact, so $z \notin M$ and all three nonempty occupied-edge complements contain z . Every point of M must lie in at least two singleton supports: with at most one, its omission count is at most $2K + 2 < t$. It must therefore lie in $R_1 \cap R_2$, and cross intersection puts it in the union of those three complements. Lemma 4.4(b), with $b = 3$, is now a contradiction.

For $(2, 1, 0)$ write

$$\mathcal{C}_{12} = \{\emptyset, D, E\}, \quad \mathcal{C}_{13} = \{\emptyset, F\}, \quad \mathcal{C}_{23} = \{\emptyset\}.$$

Again $|\mathcal{C}_1| = K - 2 \geq 8$ forces a source-1-only coordinate; the same exact count puts it outside M and in $D \cap E \cap F$. Lemma 4.5(a) gives $F \cap M = \emptyset$. Every point of M has two singleton supports, while $R_2 \cap R_3 = \emptyset$, so $M \subseteq D \cup E$. The common outside point makes D, E nested, and the sharp part of Lemma 4.5(b) says their union meets at most one deletion block. Together with $F \cap M = \emptyset$, this cannot cover the three blocks.

For $(1, 1, 1)$, Lemma 3.3 makes every pair full. Each singleton family has size $K - 1 \geq 9$, so two incident generators cannot generate it; choose a source-only coordinate at each source. The exact bound

$$4 + (K - 2) + 2 + K = t$$

puts each such coordinate outside M and in both incident pair complements. Applying Lemma 4.5(a) to each unique pair family shows that all three nonempty pair complements are disjoint from M . Yet every point of M has two singleton supports and therefore belongs, by cross intersection, to one of those complements. This is impossible and completes the $q_0 = 4$ face.

For completeness, the high-empty faces use the same hypotheses explicitly. When $(q_0, q_2) = (5, 2)$, M has four deletion blocks. In both partitions $(2, 0, 0)$ and $(1, 1, 0)$, the bound for a point of M with at most one singleton support is strictly below t ; hence every point has two supports and M is covered by the two nonempty pair complements. Part (b) of Lemma 4.5 in the concentrated case, and two applications of part (a) in the split case, permit at most two occupied blocks, not four. When $(q_0, q_2) = (6, 1)$, the same count gives $M \subseteq D$ for the unique nonempty pair complement. Projection by P_0 forces $D = M$, and then projection by each P_ℓ forces every one of the five nonempty Q_ℓ to equal M . Finally, for $(q_0, q_2) = (7, 0)$ all pair families are $\{\emptyset\}$ and the singleton supports are pairwise disjoint. For $y \in M$,

$$o(y) \leq 1 + (K - 5) + K = 2K - 4 < t = 2K - 2,$$

contradicting residual capacity. Thus boundary seven is empty.

4.3. Boundary eight. Lemma 4.6 gives $q_0 - 1 \leq q_2$, excluding $q_0 \geq 5$. The following table is the complete unordered ledger for the remaining thirty-four distributions; no row is suppressed.

q_0	pair masses	terminal mechanism
4	(4, 0, 0), (3, 1, 0), (2, 2, 0)	exact degree-four source
	(2, 1, 1)	three-block projection
3	(5, 0, 0), (4, 1, 0), (3, 2, 0)	strict degree-five source and saturation
	(3, 1, 1), (2, 2, 1)	exact degree-four source
2	(4, 2, 0), (3, 2, 1), (2, 2, 2)	exact degree-four source
	(6, 0, 0), (5, 1, 0)	effective generators and saturation
	(4, 1, 1), (3, 3, 0)	effective generators and saturation
1	(4, 3, 0), (3, 3, 1), (3, 2, 2)	exact degree-four source
	(7, 0, 0), (6, 1, 0), (5, 2, 0)	effective leaf collapse
	(5, 1, 1), (4, 2, 1)	effective leaf collapse
0	(4, 4, 0), (4, 3, 1), (4, 2, 2)	exact degree-four source
	(8, 0, 0), (7, 1, 0), (6, 2, 0)	effective leaf collapse
	(6, 1, 1), (5, 3, 0), (5, 2, 1)	effective leaf collapse
	(3, 3, 2)	Lemma 3.8

We first dispose of $q_0 = 4$. Here $t = 2K + 5$ and $|\mathcal{C}_i| = K + 2 - e_i$. In each of (4, 0, 0), (3, 1, 0), (2, 2, 0) there is a degree-four source with a full opposite zero pair. Lemma 4.1 generates its singleton family by four incident complements, and all four are effective because its size is $K - 2 \geq 8 > H(3)$. In (4, 0, 0) this makes the endpoint supports equal; an isolated-leaf point then occurs in at least $t - 4 - K = K + 1 = |\mathcal{C}_3| - 1$ leaf complements, contradicting Lemma 4.2. In (3, 1, 0) a point of the weight-three leaf outside the source has bound $2K + 3 < t$, and that leaf is then three-generated, so has size at most seven rather than $K - 1$. In (2, 2, 0) the analogous bound is $2K + 4 < t$ at either leaf, after which two generators give at most four members rather than K .

It remains on this face to treat (2, 1, 1). Write

$$\mathcal{C}_{12} = \{\emptyset, D, E\}, \quad \mathcal{C}_{13} = \{\emptyset, F\}, \quad \mathcal{C}_{23} = \{\emptyset, G\}.$$

Let M have deletion blocks Q_1, Q_2, Q_3 . A point of M with at most one singleton support has omission count below t ; in the only exact subcase, termwise saturation would make the three singleton supports pairwise disjoint and then leave a point of R_1 below capacity. Hence every point of M has at least two singleton supports and

$$(25) \quad M \subseteq D \cup E \cup F \cup G.$$

Each of sources one and two has a source-only coordinate, since otherwise three incident generators would bound its family by seven, while its size is $K - 1 \geq 9$. The safe bound for either such coordinate is $t + 1$; being in M , or missing two of the four pair complements, lowers it below t . Thus there are $z_1, z_2 \in P_0$ each lying in at least three of D, E, F, G .

If each of F, G contains one of z_1, z_2 , then Lemma 4.5(a) makes both disjoint from M , and (b) says $D \cup E$ meets at most two blocks, contrary to

(25). Otherwise, say F contains neither coordinate. Both coordinates then lie in $D \cap E \cap G$; part (a) makes G disjoint from M , while the sharp clause of part (b) lets $D \cup E$ meet at most one block. The remaining F meets at most one more. Again three blocks cannot be covered. This closes $q_0 = 4$ and also records every hypothesis of the projection lemma used in the argument.

For $q_0 = 3$, the three degree-five rows are handled as follows. A degree-five source has source-only bound $t - 1$, so its family is generated by the five incident complements; at least four are effective. The exact terminal implications are

pair masses	support consequence	terminal bound
(3, 2, 0)	both leaves lie in the source	leaf sizes at most 8 and 4
(4, 1, 0)	heavy leaf lies in the source	one-generator or equality collapse
(5, 0, 0)	endpoint supports coincide	leaf incidence $\geq \mathcal{C}_3 - 1$

In the middle row, if the light leaf is not contained in the source, its omission bound equals t ; equality makes its unique pair complement lie in every leaf member, while every source-leaf cross intersection is empty. The leaf support is then that complement and all leaf members coincide. The last row contradicts Lemma 4.2. The remaining rows (2, 2, 1) and (3, 1, 1) have degree-four sources. Exact saturation in Lemma 4.1 generates the source families by all four incident complements. In (2, 2, 1), effectiveness forces both leaves into the source, and a three-generated leaf would have at most eight rather than $K + 1$ members. In (3, 1, 1) both degree-four endpoints have all four generators effective; a coordinate outside their union has bound $2K + 5 < t = 2K + 7$, so the remaining singleton family is generated by its two weight-one complements and has at most four rather than $K + 2$ members.

For $q_0 = 2$, the exact degree-four lemma removes the first line in the ledger. For the four residual rows, the proof-critical bounds are

pair masses	strict bound	conclusion
(6, 0, 0)	isolated leaf incidence $\geq \mathcal{C}_3 - 1$	Lemma 4.2
(3, 3, 0)	leaf outside source: $2K + 7 < t$	three-generated leaf, size ≤ 7
(5, 1, 0)	heavy leaf outside source: $< t$	one-generator/equality collapse
(4, 1, 1)	outside two sources: $2K + 8 < t$	two-generated leaf, size ≤ 4

The equality collapse in (5, 1, 0) is the same cross-intersection argument as above: equality forces the unique light-edge complement into every leaf member and every source-leaf cross intersection to be empty, so all leaf members coincide. In (3, 3, 0) the imported pair dichotomy only supplies the full/nonfull alternatives on the weight-three edge; its fullness follows from the project exact-threshold Lemma 3.3.

For $q_0 = 1$, the exact degree-four rows are the three listed in the ledger. In the remaining five rows, source inclusion and the number of inactive generators give the following strict sequence; the two numbers in the last

column are the successive deficits below t when two support inclusions are needed.

pair masses	terminal leaf generators	omission deficits
(7, 0, 0)	Lemma 4.2	1
(6, 1, 0)	1	5, 1
(5, 2, 0)	2	4, 2
(5, 1, 1)	2	2
(4, 2, 1)	3	4

Thus the terminal singleton family has at most 2, 4, or 8 members, whereas its exact size is at least $K + 5 \geq 15$; in the concentrated row Lemma 4.2 contradicts size $K + 8$.

Finally take $q_0 = 0$. The exact degree-four lemma removes the three rows shown in the ledger. The other six nonterminal rows have the following literal omission calculations; each strict inequality first forces the displayed leaf support into the indicated source union, after which the listed number of generators gives the contradiction.

pair masses	first bound	second bound	generators
(8, 0, 0)	Lemma 4.2	—	—
(7, 1, 0)	$2K + 6$	$2K + 12$	1
(6, 2, 0)	$2K + 7$	$2K + 11$	2
(6, 1, 1)	$2K + 12$	—	2
(5, 3, 0)	$2K + 8$	$2K + 10$	3
(5, 2, 1)	$2K + 10$	—	3

Here $t = 2K + 13$, so every displayed value is strict. The resulting union bounds 2, 4, 8 are all below the exact leaf sizes. In (8, 0, 0) at least five of the eight occupied-edge generators are effective at each endpoint; an isolated-leaf coordinate consequently lies in every leaf complement, contradicting distinctness. These arguments leave only

$$(26) \quad q_0 = 0, \quad (q_{12}, q_{13}, q_{23}) = (3, 3, 2).$$

For (26), all private pairs are full and

$$t = 2K + 13, \quad |\mathcal{C}_1| = K + 4, \quad |\mathcal{C}_2| = |\mathcal{C}_3| = K + 5.$$

The degree-five source inclusions make $\mathcal{C}_2, \mathcal{C}_3$ generated by their five incident pair complements. The height-three generator bound gives $K \in \{10, 11\}$. Write

$$\mathcal{C}_{12}^+ = \{A_1, A_2, A_3\}, \quad \mathcal{C}_{13}^+ = \{B_1, B_2, B_3\}, \quad \mathcal{C}_{23}^+ = \{D_1, D_2\}.$$

Lemma 3.8, applied at sources two and three, makes each list of five generators pairwise disjoint and actual singleton-family members. Cross intersection gives $A_i \cap B_j \in \{\emptyset, D_1, D_2\}$; disjointness from D_1, D_2 forces this intersection to be empty. Thus all eight pair complements are pairwise disjoint. For $z \in A_1$, the three singleton contributions are at most $K + 4, 5, 0$,

the pair contribution is one, and the full-support contribution is at most K . Therefore

$$o(z) \leq (K + 4) + 5 + 1 + K = 2K + 10 < t,$$

a contradiction. Boundary eight is empty.

Proof of Theorem 1.8. Proposition 1.7 excludes $q_0 + q_2 \leq 5$. The three subsections above exclude equality at six, seven, and eight. Hence $q_0 + q_2 \geq 9$. $\square$

5. DOMINANCE AND THE REMAINING INCIDENCE CAPACITY

On R , define the dominance order

$$y \preceq z \iff (y \in F \implies z \in F) \text{ for every } F \in \mathcal{F}.$$

Separation makes this a partial order. Let C_z be the union of all members avoiding z . Then

$$(27) \quad y \in C_z \iff y \not\preceq z.$$

Lemma 5.1 (Core capacity). *Fix $q = q_0$ and a block pattern (4). After the q empty-support members, the r maximal avoiders C_z , and the seven Boolean joins with M are removed, the total incidence still available on the coordinates of M is at most*

$$(28) \quad \mathcal{C} = (t - q - r - 7)m + S + E = (r + s - q - 2)m + S + E,$$

where

$$(29) \quad E = \sum_i \frac{d_i(d_i + 1)}{2} + hS + \frac{h(h + 1)}{2}.$$

Proof. The empty-support members contribute $qm - S$ incidences on M . If $y \in D_i$, the witnesses $M \setminus D_j$ for $j \neq i$ and the maximum M imply

$$\uparrow y \subseteq D_i \cup H,$$

where H is the uncovered part. If $y \in H$, then $\uparrow y \subseteq H$. Antisymmetry bounds the total number of ordered comparable pairs internal to a k -point part, including the diagonal, by $k(k + 1)/2$. Allowing every direction from a block to H gives

$$\sum_{y \in M} |\uparrow y| \leq E.$$

By (27), the r maximal avoiders therefore contribute at least $rm - E$ incidences on M . The seven joins of M with the nonempty Boolean members contribute another $7m$. These collections are disjoint: their private supports differ, except that the full-support Boolean join is U , whereas every C_z omits z . Since every coordinate has frequency at most t , subtraction from the total capacity tm gives (28). $\square$

6. THE BLOCK-PROJECTION RECURRENCE

Fix one of the six proper nonempty fibers. Remove the member obtained by joining its Boolean member with M ; if the fiber count is L , exactly L members remain. For a residual trace T , pass to the complement

$$E = M \setminus (T \cap M).$$

Stratify the complements by their common trace on $R \setminus M$.

Proposition 6.1 (Exact projection closure). *Before the single full- M trace is removed, the complement family in each stratum is intersection-closed and satisfies*

$$(30) \quad E \mapsto E \cap D_i \quad (1 \leq i \leq q-1).$$

Every ordinary retained stratum contains the empty complement and has both properties. In the unique special stratum, a nonempty projection $E \cap D_i$ is retained; the empty projection alone may be the removed complement. In particular, the retained special stratum need not itself be intersection-closed.

Proof. Before removal, union of two traces corresponds to intersection of their complements. Moreover,

$$M \setminus (T \cup (M \setminus D_i)) = (M \setminus T) \cap D_i.$$

Joining with M produces the empty complement without changing the trace on $R \setminus M$. Exactly one such trace is removed. All nonempty projected complements therefore remain in the special stratum, while a projection equal to the empty complement may be the deleted member. $\square$

For $0 \leq a \leq r$ and $0 \leq n \leq 2^a$, let $P_a(n)$ be the sum of the n largest cardinalities among all subsets of an a -set. For $d \geq 1$ and $0 \leq n \leq 2^d - 1$, let $R_d(n)$ be the analogous sum over nonempty subsets of a d -set. Both functions are explicit binomial sums.

For a stratum containing k remaining complements, put $\varepsilon = 1$ if the empty complement is present and $\varepsilon = 0$ otherwise. Define

$$(31) \quad \begin{aligned} \Psi_\varepsilon(k) = \max_{(t_i)} & \left[\sum_i R_{d_i}(t_i) \right. \\ & \left. + P_{h+\sum_{i:t_i>0} d_i} \left(k - \varepsilon - \sum_i t_i \right) \right], \end{aligned}$$

where

$$0 \leq t_i \leq 2^{d_i} - 1, \quad N := k - \varepsilon - \sum_i t_i \in \left[0, 2^{h+\sum_{i:t_i>0} d_i} \right],$$

the maximum being over precisely these admissible integer tuples, and set $\Psi_0(0) = 0$. Define

$$(32) \quad \Omega(0) = 0, \quad \Omega(n) = \max_{1 \leq k \leq n} \{ \Omega(n-k) + \Psi_1(k) \}.$$

and

$$(33) \quad \Theta_{d,h}(L) = \max_{0 \leq k \leq L} \{\Psi_0(k) + \Omega(L - k)\}.$$

Lemma 6.2 (Block-projection bound). *The total incidence on M of the L remaining traces is at least*

$$(34) \quad mL - \Theta_{d,h}(L).$$

Proof. In one stratum, let t_i be the number of distinct nonempty complements contained in D_i . Their total cardinality is at most $R_{d_i}(t_i)$. Every other complement has each nonempty projection onto D_i among those local complements by Proposition 6.1. It therefore avoids D_i when $t_i = 0$ and otherwise lies in the union of the uncovered part and the active blocks. The nonlocal complements are distinct, so their total cardinality is at most the corresponding P -term in (31). Notice that allowing this P -term to reuse empty or local subsets only enlarges the upper bound, so no subtraction assumption is hidden here.

Equation (32) ranges over every ordered partition into ordinary strata. Equation (33) then adds at most one exceptional stratum. Thus Θ is an upper bound on complement incidence. Subtracting from mL proves (34). $\square$

The block formulation is necessary. A non-singleton block cannot generally be split into coordinate generators.

Proposition 6.3 (Sharp local obstruction). *For every $d \geq 2$ and $L \geq 1$, there is a height-at-most-three union-closed trace family with a block D of size d such that (30) holds, while the L traces remaining after removal of the full trace have total incidence zero on D .*

Proof. Take disjoint sets D, Z with $|D| = d$ and $|Z| = L - 1$, put $M = D \cup Z$, and use

$$\mathcal{T} = \{M, Z\} \cup \{Z \setminus \{z\} : z \in Z\}.$$

Its complement family is

$$\{\emptyset, D\} \cup \{D \cup \{z\} : z \in Z\}.$$

It is intersection-closed and closed under intersection with D . After M is removed, every trace is disjoint from D . The longest strict chain has at most three members. $\square$

7. A UNIFORM ANALYTIC BLOCK BOUND

Put $b = q_0 - 1$. For integers $b \geq 1$ and $L \geq 0$, define

$$\phi_b(L) = bL - \max_{0 \leq a \leq \min(b, L)} a(L + 1 - a).$$

Equivalently,

$$(35) \quad \phi_b(L) = \begin{cases} bL - \lfloor (L + 1)^2 / 4 \rfloor, & L \leq 2b - 1, \\ b(b - 1), & L \geq 2b - 1. \end{cases}$$

Its discrete increments are

$$(36) \quad \phi_b(L) - \phi_b(L-1) = \max\{0, b - \lceil L/2 \rceil\}.$$

Thus ϕ_b is nondecreasing and discretely concave, every increment is at most b , and

$$(37) \quad \phi_b(a) + \phi_b(c) \geq \phi_b(a+c).$$

Lemma 7.1 (Ground-independent block bound). *Suppose $q_0 = q \geq 2$, and fix one proper nonempty fiber with L residual rows after its Boolean join with M is removed. Its total incidence on M is at least $\phi_{q-1}(L)$.*

Proof. Consider one complement stratum. Let a blocks be active, let t_i be the number of distinct nonempty local complements contained in active block D_i , put $T = \sum t_i$, and let $N = n - T$, where n is the number of nonempty complements in the stratum. Thus $T \geq a$. Distinctness gives

$$\sum \{|E| : \emptyset \neq E \subseteq D_i, E \text{ local}\} \leq t_i d_i - (t_i - 1),$$

because at most one local set is the full block. By Proposition 6.1, every nonlocal complement projects to one of these local complements on each active block and is empty on each inactive block. Hence its size is at most $h + \sum_{i \text{ active}} d_i$.

Let $\varepsilon = 1$ for an ordinary stratum and 0 for the special stratum. Subtracting the resulting complement bound from full trace incidence yields

$$\varepsilon m + Tb - a + N(b - a).$$

For fixed a, n this is minimized at $T = a$ and becomes

$$\varepsilon m + bn - a(n + 1 - a) \geq \varepsilon m + \phi_b(n).$$

This argument retains each true block; no nonsingleton block is split into coordinate generators.

Sum over all strata. Equation (37) combines the nonempty rows. Every ordinary empty-complement row contributes m , while adding one argument to ϕ_b costs at most $b \leq m$ by (36) and Lemma 2.2. Inserting all such rows therefore leaves at least $\phi_b(L)$. $\square$

Lemma 7.2 (Six-fiber lower bound). *Let $q = q_0 \geq 3$ and*

$$U = \lfloor (q + s)/2 \rfloor.$$

The total incidence on M of the six proper nonempty fibers is at least

$$(38) \quad \mathcal{I}_*(r, s, q) = \min_{u \in \{0, U\}} (2\phi_{q-1}(r + 1 + s - u) + \phi_{q-1}(r + 1 - q + u)).$$

Proof. For the actual value of u , Lemma 2.1 says that all six fiber counts are at most w , the singleton counts are at least ℓ , and their sum is $2w + \ell$. The vector

$$(w, w, \ell, 0, 0, 0)$$

majorizes every such six-vector. Concavity of ϕ_{q-1} and Lemma 7.1 therefore give

$$2\phi_{q-1}(w) + \phi_{q-1}(\ell).$$

Nonnegativity of q_2 gives $0 \leq u \leq U$. The displayed expression is discretely concave in u , so its minimum on this enlarged interval occurs at an endpoint. Including an infeasible endpoint can only weaken the bound. $\square$

Lemma 7.3 (Capacity endpoints). *Let $q = q_0 \geq 3$. The capacity in Lemma 5.1 is at most*

$$(39) \quad \mathcal{C}_*(r, s, q) = \max\{C_0, C_r\},$$

where, with $b = q - 1$,

$$(40) \quad C_0 = b(r + s - q),$$

$$(41) \quad C_r = (r + s - q - 2)r + 2b + (r - b)b + \frac{(r - b)(r - b + 1)}{2}.$$

Proof. Write $d_i = 1 + x_i$, $X = \sum x_i$, $m = b + e$, and $h = e - X$. For fixed e , the quantity $S + E$ in Lemma 5.1 is at most its value when all blocks are singletons and all excess points are uncovered. Indeed, direct subtraction gives

$$\left(2b + eb + \frac{e(e+1)}{2}\right) - (S + E) = \frac{X^2 - \sum_i x_i^2 + 2X(b-2)}{2} \geq 0.$$

The resulting capacity is a convex quadratic in $0 \leq e \leq r - b$, so its maximum occurs at $e = 0$ or $e = r - b$, giving (40) and (41). $\square$

We obtain the following explicit criterion, useful beyond the particular range isolated below:

$$(42) \quad \mathcal{I}_*(r, s, q) > \mathcal{C}_*(r, s, q) \implies q_0 \neq q.$$

Proof of Theorem 1.2. Assume $q \geq \lceil (2r + s)/3 \rceil$. Put

$$p = r - q, \quad \delta = q - 2p - s.$$

Then

$$(43) \quad q \geq 7, \quad \max(-1, 10 - q) \leq p \leq q/2, \quad 0 \leq \delta \leq q - 2p.$$

Moreover, $C_r \geq C_0$, since

$$C_r - C_0 = \frac{(q - r - 1)(q - 3r - 2s + 4)}{2} \geq 0,$$

and

$$(44) \quad C_r = q^2 + q - 2 - \delta(q + p) - \frac{p^2}{2} + pq - \frac{3p}{2}.$$

It remains to check the two values of u in (38). We repeatedly use the globally valid consequence of (35)

$$(45) \quad \phi_{q-1}(L) \geq (q - 1)L - \frac{(L + 1)^2}{4}.$$

For $u = 0$, the lower gap furnished by (45) is concave in δ , so only $\delta = 0$ and $\delta = q - 2p$ are needed. Four times these two gaps are

$$\begin{aligned} A &= -p^2 + 14p + 4q^2 - 24q - 16, \\ B &= -9p^2 - 18p + 6q^2 - 8q - 16. \end{aligned}$$

Both are concave in p . At $p = q/2$ both endpoint bounds reduce to $(3q - 16)(5q + 4)/4 > 0$. At $p = -1$, relevant when $q \geq 11$, they are $4q^2 - 24q - 31$ and $6q^2 - 8q - 7$. At the remaining lower endpoint $p = 10 - q$, where $7 \leq q \leq 11$, the first is $3(q - 4)(q - 2)$ and the second is at least 87. Thus the $u = 0$ value exceeds C_r .

For $u = U$, write $\delta = 2k$ or $2k + 1$. Four times the corresponding lower gaps are

$$\begin{aligned} E(k) &= -3k^2 + (8p + 2q + 24)k + 2p^2 - 4pq + 6p + 5q^2 - 16q - 16, \\ O(k) &= -3k^2 + (8p + 2q + 22)k + 2p^2 - 4pq + 10p + 5q^2 - 14q - 9. \end{aligned}$$

They are concave in k . At $k = 0$, their values at the minimizing endpoint $p = q/2$ are

$$\frac{7q^2 - 26q - 32}{2}, \quad \frac{7q^2 - 18q - 18}{2},$$

both positive for $q \geq 7$. At the last k of each parity, the four possible numerators, after multiplication by 4, have endpoint values

	$p = -1$	$p = q/2$
1	$21q^2 - 20q - 28$	$2(7q^2 - 26q - 32)$
2	$21q^2 - 12q - 68$	$2(7q^2 - 30q - 68)$
3	$3(7q^2 - 6q - 17)$	$14q^2 - 64q - 115$
4	$21q^2 - 14q - 43$	$14q^2 - 48q - 83$

All are positive for $q \geq 7$. These numerators are concave in p , so the table handles $q \geq 11$. For $7 \leq q \leq 10$, substitution of $p = 10 - q$ gives respective minima 397, 285, 276, 408. The $u = U$ value also exceeds C_r . Thus (42) excludes every $q \geq \lceil (2r + s)/3 \rceil$, as claimed. $\square$

8. EXACT EXCLUSION AT $(r, s) = (10, 2)$

We now specialize to $r = 10$ and $s = 2$ and prove Theorem 1.3. Fix $q \in \{5, \dots, 11\}$. By Lemma 2.2, every possible empty-support fiber is represented by

$$(46) \quad d_1 + \dots + d_{q-1} \leq m \leq 10, \quad d_i \geq 1, \quad h = m - \sum_i d_i.$$

For $0 \leq u \leq \lfloor (q + 2)/2 \rfloor$, enumerate the nonnegative triples

$$x_{12} + x_{13} + x_{23} = q + 2 - 2u$$

and the six fiber counts in Lemma 2.1. For a vector $\ell = (\ell_1, \dots, \ell_6)$, Lemma 6.2 gives the incidence lower bound

$$(47) \quad m \sum_{j=1}^6 \ell_j - \sum_{j=1}^6 \Theta_{d,h}(\ell_j).$$

Subtract the capacity (28). Exact evaluation gives Table 1.

TABLE 1. Minimum exact incidence gaps.

q	block patterns	vectors	minimum gap	minimizing $(m; d_i; h; u)$
5	71	4970	10	(10; 1, 1, 1, 7; 0; 3)
6	45	4275	39	(5; 1, 1, 1, 1, 1; 0; 4)
7	26	3250	67	(6; 1, 1, 1, 1, 1, 1; 0; 4)
8	14	2254	86	(7; 1, 1, 1, 1, 1, 1, 1; 0; 5)
9	7	1421	113	(8; 1, 1, 1, 1, 1, 1, 1, 1; 0; 5)
10	3	756	129	(9; 1, 1, 1, 1, 1, 1, 1, 1, 1; 0; 6)
11	1	308	153	(10; 1, 1, 1, 1, 1, 1, 1, 1, 1, 1; 0; 6)

Every gap is positive, contradicting Lemma 5.1. Hence none of $q_0 = 5, \dots, 11$ is possible. Since $q_0 \leq 11$, Theorem 1.3 follows.

Remark 8.1. The same safe relaxation has minimum gap -30 at $q_0 = 4$. Thus the recurrence itself does not justify a further descent. Any improvement must use additional compatibility between fibers or the projection-matching conditions from [6].

9. EXACT CERTIFICATE AND REPRODUCIBILITY

The certificates use only the Python standard library and integer arithmetic. The source archive contains the following standalone checkers:

- (i) the primary fixed-point checker evaluates (31) by merging states with the same local count and active-block size, and evaluates (32) by dynamic programming;
- (ii) an independent fixed-point checker directly enumerates every local-count tuple and every ordered composition of the ordinary strata;
- (iii) the primary analytic checker evaluates (38)–(39) directly and retains the full u interval as a check on the endpoint reduction;
- (iv) an independent analytic checker shares no imports with the preceding program. It exhausts 1,112,048 small projection-closed strata (without assuming intersection closure), 1,064 capacity slices, and 6,573 six-fiber dynamic programs;
- (v) the primary low-end checker enumerates every witness type and occupied pair support when $q_0 + q_2 \leq 1$, and evaluates the terminal incidence constraints exactly;

- (vi) an independent declarative checker imports no project code and exhausts the corresponding witness-allocation and coordinate-bit truth tables. It checks nine zero-pair witness rows, three global one-pair allocations, and twenty-four terminal truth rows;
- (vii) the primary boundary checker exhausts all ten distributions of two units of pair-fiber mass and the admissible sets of five-rows;
- (viii) an independent boundary checker rederives the private-defect table and exhausts the terminal residual-incidence truth tables for the two-high $(1, 1)$ and $(2, 0)$ cases. Together the two scripts reproduce the high-row caps and the finite trace-model counts used as regression data for the displayed classification proof.
- (ix) the exact-threshold checker verifies the parity-split three-trace arithmetic, every boundary identity, and the terminal low-row defect table; an independent checker rebuilds the same assertions from a declarative truth-table scan without importing project code.
- (x) the primary $q_0 + q_2 = 3$ checker verifies the exact fiber sizes, omission bounds, and minimum support-zero matching counts in all four boundary types. An independent affine-coefficient checker and a weighted-edge truth table share no project imports. They verify the arithmetic and the three-edge allocation table. The cross-fiber contradiction itself is the human argument in the preceding section.
- (xi) four pairs of face regression checkers audit the cases

$$(q_0, q_2) = (4, 0), (3, 1), (2, 2), (1, 3).$$

For each face, one implementation replays the affine omission and fiber-size identities over a redundant integer grid. The other tests the selected pointwise intersection or generator-union implications on independently encoded finite set systems. The $(2, 2)$ scripts enumerate both pair-weight partitions and the full/nonfull concentrated split. These programs do not encode the three-cover hypotheses end to end and do not constitute a separate exhaustive proof of a face.

- (xii) the primary $(0, 4)$ regression checker enumerates all four partitions of pair mass four and checks every displayed affine capacity identity over its stated grid. An independent implementation checks the symbolic degree identities and 28,576 separately encoded bit-set/generator rows. The arbitrary-parameter and arbitrary-ground conclusions are the displayed human reductions, except for the proof-essential finite lemma in the next item.
- (xiii) the primary four-generator checker exhausts all 32,768 Venn-atom presence patterns, retains 301 labelled intersection-closed realizations, and finds balanced cross-meet bound eight. The independent checker starts from the 5^6 abstract pair-meet tables, finds fifteen associative meet types, recovers the same 301 realizations, and independently returns eight.

- (xiv) five pairs of boundary-five regression checkers audit the faces with $q_0 \geq 1$ and the reduction of the $q_0 = 0$ face. The primary programs enumerate unordered pair-mass partitions, threshold positions, affine fiber sizes, and omission gaps. The independent programs enumerate all labelled edge allocations and separately encoded pointwise generator implications. These programs audit the displayed human algebra and do not extrapolate from a bounded residual ground.
- (xv) the proof-essential primary five-generator checker first enumerates 202 disjoint atom families in B_5 , constructs all 15,066 height-three intersection families, retains the 711 of size at least twelve, enumerates 12,656 admissible six-element meet families, and rejects all 253,116 unordered large-family pairs. It contains a fail-closed assertion that aborts on any compatible witness. The independent checker starts from all 7,580 nonzero Boolean antichains, reconstructs the same finite domains without importing the primary code, and again finds no witness.
- (xvi) the boundary-six runner replays eighteen checks in nine independent pairs. They audit the source-degree ledger, high-empty faces, the complete $q_0 = 4$ face, the concentrated five-generator endpoint, the split $q_0 = 2$ face, the balanced $q_0 = 3$ equality case, both nonconcentrated $q_0 = 0$ rows, and the final concentrated $q_0 = 1$ and $q_0 = 0$ rows.
- (xvii) the boundary-seven runner replays fourteen checks in seven independent pairs: one pair for the concentrated endpoint reduction, one for each face $q_0 = 0, 1, 2, 3, 4$, and one for the combined high-empty faces $q_0 = 5, 6, 7$. The scripts audit partition completeness, affine gaps, effective-generator minima and the small projection truth tables.
- (xviii) the boundary-eight fail-closed runner

`run_qsum8_combined_frontier.py`

executes twenty checks in ten independent pairs. They cover, respectively: the deletion-block injection; degree-five equality capacity; exact degree-four source saturation; the $q_0 = 4$ degree-four rows; the last $q_0 = 4$ projection row; the complete $q_0 = 2$ and $q_0 = 1$ faces; the first six $q_0 = 0$ rows; the near-extremal five-generator classification; and the final $(3, 3, 2)$ omission contradiction. Each primary checker and its independent counterpart use different data generation or truth-table encodings. The runner stops on a nonzero exit code or a missing terminal `RESULT PASS` marker and reports exactly twenty passes before printing the boundary-eight exclusion marker; it does not by itself certify the lower boundaries.

- (xix) the publication entry point `run_qsum9_source_certificates.py` first replays the forty-two earlier manuscript checks and then the boundary-six, boundary-seven and boundary-eight runners in that

order. Only after seeing their exact terminal markers does it print `THEOREM CHAIN q0+q2>=9`. Every release runner refuses optimized Python mode, and each packaged checker fails immediately when `--debug--` is false, so proof-critical assertions cannot silently disappear under `python -O`.

The first two programs independently return the seven gaps in Table 1. The finite loop ranges are exactly:

1. $q = 5, \dots, 11$;
2. $q - 1 \leq m \leq 10$;
3. every nondecreasing positive integer partition $d_1 + \dots + d_{q-1} \leq m$;
4. $0 \leq u \leq \lfloor (q+2)/2 \rfloor$;
5. every nonnegative triple summing to $q+2-2u$;
6. every local count $0 \leq t_i \leq \min(2^{d_i}-1, k-\varepsilon)$, subject to the displayed distinct-subset admissibility bound for N ;
7. every ordered composition of at most thirteen remaining rows.

There is no numerical optimization, tolerance, random seed, SAT assumption, or finite-to-infinite extrapolation. The infinite conclusion in Theorem 1.2 rests on the algebra in Section 7, not on a finite grid. The fixed-point programs verify only the finite arithmetic after Lemmas 5.1 and Lemma 6.2; those lemmas are proved for arbitrary trace layers. Likewise, Theorem 1.4 is the set-theoretic argument in Section 3; its two finite checkers are independent regression audits of the exhaustive terminal case split, not finite-ground proofs. The same separation applies to Theorem 1.5: the two boundary checkers audit its integer and Boolean case tables, while the cross-fiber trace implications are proved in the text. The exact-threshold programs similarly audit only the integer identities in Lemma 3.3 and the final defect table; the parity argument and projection contradiction are the displayed human proof. The two $q_0 + q_2 = 3$ programs audit the affine arithmetic and finite three-edge allocation only. The face programs for $q_0 + q_2 = 4$ likewise audit exact identities and finite intersection implications; the ground-independent complement and omission arguments proving the boundary-four part of Theorem 1.7 are displayed above. The boundary-five face programs likewise audit exact identities and finite edge allocations, while the general height-three generator bound is proved in Lemma 3.6. The finite lemma used in the boundary-four proof is Lemma 3.5. Its finite search is exhaustive for arbitrary ground sets because it first reduces every coordinate to one of fifteen Venn membership words; it is not a bounded-ground extrapolation. The additional proof-essential Lemma 3.7 is exhaustive for arbitrary ground sets because the injective intersection-preserving map σ reduces every five-generator instance exactly to B_5 . Its two programs use independent family parametrizations and both fail closed on a witness. The twenty boundary-eight programs have the same limited role: they audit the complete integer partition ledger, the equality/strict omission arithmetic, and the finite

Boolean or Venn images after the human reductions in Section 4. The universal source, projection, generation, and deletion-block arguments are proved in that section. In particular, the programs do not search bounded residual ground sets and do not infer Theorem 1.8 from finite experiments.

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The conditional three-cover structure used here is due to Tian [6]. The results in Theorems 1.2 and 1.3, together with Theorems 1.4, 1.5, 1.6, Proposition 1.7, and Theorem 1.8 constrain one branch of that structure and leave the general height-five problem open. The boundaries $q_0 + q_2 \leq 8$ are excluded only in the full minimal three-cover branch. Other three-cover configurations and both four-cover configurations in Tian’s reduction remain unresolved.

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